University of Otago

ŌTĀKOU WHAKAIHU WAKA

A Protein First Computational Screen for Structured RNA Elements

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Abstract

RNA underpins many of the most transformative advances in modern biotechnology, including RNA interference, CRISPR systems, metabolite-sensing riboswitches, and recently discovered bridge RNA-guided recombination systems. Despite their importance, discovering new functional RNAs remains challenging. Unlike proteins, RNA sequences often evolve rapidly and are conserved primarily through structure rather than sequence, limiting the effectiveness of conventional homology-search methods. Protein homology detection in contrast is an easier solved problem with robust and fast implementations.

Here, we present a protein-first RNA discovery framework that leverages conserved bacterial protein domains to identify candidate cis-regulatory RNAs within neighbouring non-coding regions. Minimum free energy folding is unreliable for single sequences and machine learning structure prediction, like AlphaFold, struggle with the small alphabet and the limited training data available. Rather than relying solely on structure prediction, we use comparative genomics and identify covarying sites, which mutate together across evolution to preserve base pairing, as evidence of real structure. As an initial application, we focus on light-associated protein systems to search for novel light-responsive RNA regulators - looking for potential RNA light switches. We further introduce an approach for measuring relatedness between RNA elements, filling a gap in this area. Together, this framework provides a targeted and scalable route to RNA discovery, allowing biological pathways of interest to be rapidly screened for previously uncharacterized functional RNA elements.

Acknowledgements

Firstly I would like to thank my supervisor Paul Gardner for his guidance but equally for granting me the freedom to steer this project where it went. We often talked about how at times, with the endless possibilities in a project like this, it was difficult making it through the weeds. As such I am particuarly grateful for the direction back toward the garden path.

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Abbreviations

MSA	Multiple Sequence Alignment	16
HMM	Hidden Markov Model	iv
CM	Covariance Model	20
GTDB	Genome Taxonomy Database	8
BLAST	Basic Local Alignment Search Tool	4
NCBI	National Center for Biotechnology Information	13
COVID-19	Coronavirus Disease 2019	2
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats	viii
crRNA	CRISPR RNA	2
DMS-MaPseq	Dimethyl Sulfate Mutational Profiling with Sequencing	6
HDV	Hepatitis Delta Virus	2
mRNA	Messenger RNA	2
RNAi	RNA Interference	2
RNP	Ribonucleoprotein	v
rRNA	Ribosomal RNA	2
SELEX	Systematic Evolution of Ligands by Exponential Enrichment	6
SHAPE-seq	Selective 2'-Hydroxyl Acylation analysed by Primer Extension Sequencing	6
sRNA	Small Regulatory RNA	2
KS	Kolmogorov-Smirnov	26
BLUF	Blue-Light Using FAD	v
FAD	Flavin Adenine Dinucleotide	v
iTOL	interactive Tree Of Life	16
FMN	Flavin Mononucleotide	v
SAM	S-adenosylmethionine	29
Gb	Gigabyte	20
Mb	Megabyte	20
CPU	Central Processing Unit	20

Chapter 1

Introduction

RNAs play a wide range of roles in cellular processes, many of which have been harnessed for biotechnological applications, and are likely to have many more functions yet to be discovered. This project aims to identify novel conserved RNA structural elements associated with light-sensing gene families in bacteria. This introduction provides background and rationale for the screening of novel functional RNA motifs in bacterial genomes, and outlines the aims and significance of the project.

1.1 Ribonucleic Acids

The central dogma of molecular biology, first articulated by Francis Crick, describes the flow of genetic information: DNA $\rightarrow$ RNA $\rightarrow$ Protein[1]. In this classical view, RNA serves primarily as an intermediate messenger, transcribing genetic information from DNA and translating it into proteins at the ribosome. However, this perspective underrepresents the diversity in the roles of RNA molecules.

In the hypothetical RNA world, RNA molecules are thought to have performed many of the roles now largely carried out by proteins and DNA, including catalysis, regulation, and information storage[2–5]. Although these functions have been largely superseded by proteins and DNA, numerous RNA-based regulatory mechanisms persist in modern organisms.

1.2 Functional RNAs in Biology and Biotechnology

The diverse functional roles of RNA are driven by their structural and reactive diversity. Naturally occurring functional RNAs include purely RNA systems, riboswitches and ribozymes, and those in complexes with proteins, while *in vitro* selected RNAs include

fluorogens and aptazymes[6].

Ribozymes are catalytic RNAs that accelerate chemical reactions, much like protein enzymes. Beyond the ribosome and spliceosome, several small self-cleaving ribozymes exist in nature, including the hammerhead, hairpin, Hepatitis Delta Virus (HDV), glmS, twister, and pistol ribozymes[6–8].

RNA has been central to several major biotechnological breakthroughs. CRISPR-Cas9 gene editing harnesses guide RNAs for programmable genome modification Jinek et al.[9]. Messenger RNA (mRNA) vaccines against Coronavirus Disease 2019 (COVID-19) demonstrated the therapeutic potential of synthetic mRNAs[10]. RNA Interference (RNAi) enables targeted gene silencing in research and therapeutics[11].

RNA aptamers such as Spinach bind fluorogenic compounds and fluoresce upon binding, enabling RNA visualization in living cells[12]. The photoriboswitch Werewolf-1, generated through *in vitro* selection, responds to light by binding photoisomerizable ligands, offering potential for optogenetic control of gene expression[13]. These RNAs, involved in fluorescence and photoreception, have broad relevance for medical research, biotechnology, and agricultural applications[6].

Ribonucleoprotein Complexes

Make sure to introduce RNPs vaguely enough to include riboswitches While some RNAs function independently, many operate in RNP complexes. The ribosome is one such RNP with critical function in all cells. This large complex catalyses peptide bond formation, with Ribosomal RNA (rRNA) forming the catalytic core, while ribosomal proteins provide structural support and regulatory functionality[14]. Spliceosomes are another example of essential RNPs in eukaryotes[14].

In bacteria, several classes of RNPs mediate post-transcriptional regulation. Small Regulatory RNAs (sRNAs) often require the RNA chaperone Hfq to facilitate base pairing with target mRNAs, thereby modulating transcript stability[15].

CRISPR-Cas systems exemplify the complexity of bacterial RNPs. CRISPR RNAs (crRNAs) guide Cas nucleases to complementary DNA or RNA targets, enabling sequence-specific nucleic acid cleavage[16]. This mechanism has famously been repurposed as a gene-editing tool[9].

BRIDGE RNAS!!!

Riboswitches

Riboswitches are structured RNA domains that regulate gene expression in response to some stimuli, often the presence of a metabolite[17]. Although seemingly less abun-

dant than protein factors that perform similar functions, riboswitches are recognized as important regulatory elements across all domains of life. They are typically embedded within messenger RNAs and control gene expression through ligand-induced conformational changes.

Riboswitch mechanisms vary widely. Figure **??** illustrates several experimentally validated regulatory strategies[18].

Transcriptional riboswitches control formation of Rho-independent terminators—intrinsic termination structures consisting of a hairpin followed by a uridine-rich tract. Depending on the riboswitch, ligand binding may promote terminator formation or stabilize an antiterminator structure, enabling transcriptional read-through. Similarly, translational riboswitches can sequester or expose the ribosome-binding site in response to ligand binding[17].

1.3 RNA Structure Determines Function

MAKE IT CLEAR THAT STRUCTURE IS CONSERVED OVER SEQUENCE IN A LOT, BUT NOT ALL CASES, BUT HERE WE ARE FOCUSING ON STRUCTURED RNAS. INTRODUCE THE IDEA OF COVARIATION AND HOW IT IS USED TO DETECT FUNCTIONAL RNAs.

RNA molecules, much like the polypeptides that make up proteins, fold into complex structures that enable diverse functions[19]. They typically exist as single-stranded molecules, which can fold back on themselves through intramolecular base pairing to form secondary and tertiary structures.

RNA secondary structure arises from Watson-Crick base pairing (A-U and G-C) and wobble pairing between nucleotides within the same strand[20]. Common secondary structures include hairpins, consisting of a base-paired stem capped by an unpaired loop; internal loops and bulges, made up of unpaired nucleotides within a helical stem; junctions where three or more helical stems converge; and pseudoknots, where nucleotides in a loop pair with regions outside that loop[21].

Beyond secondary structure, RNAs fold into three-dimensional structures through long range interactions[22]. These include base triplets, ribose zippers, A-minor motifs, and tetraloop-receptor interactions, which stabilize compact RNA folds.

The relationship between RNA structure and function is exemplified by riboswitches, a class of regulatory RNAs. The aptamer domain folds into a specific three-dimensional pocket that selectively binds a target ligand through a combination of electrostatic interactions and shape complementarity[18]. Ligand binding stabilizes one conformational state over another, thereby modulating transcription termination or initiation. This

allosteric mechanism, whereby binding at an aptamer site affects a distal functional region, demonstrates how RNA structure enables regulatory function.

1.4 The Twilight Zone of RNA Homology

RNA homology searches are complicated by the fact that RNA structure is often more conserved than sequence, particularly in functional RNAs maybe cite[18].

1.5 The Twilight Zone of Protein Homology

Much lower... more reliable for identifying deeper homology than RNA

1.6 Current Approaches to Functional RNA Discovery

It is predicted that a large number of riboswitches remain undiscovered, potentially on the order of thousands[18]. Ronald Breaker and colleagues have pioneered systematic riboswitch and ribozyme discovery, identifying over 55 distinct riboswitch classes to date[23]. Other major RNA-related discoveries, such as ribosomes and CRISPR-Cas systems, occurred through more exploratory research approaches[24].

Computational Approaches

Bioinformatic sequence-based searches using tools like Basic Local Alignment Search Tool (BLAST) identify homologous sequences, but many functional RNAs have divergent sequences detectable only through structural conservation. Structure based searches have proven successful, like the bioinformatic search for twister ribozymes by Roth et al.[7] discovering roughly 2,700 ribozymes. These techniques search on sequence and secondary structure patterns, but require existing alignments and structure models, limiting their utility for discovering entirely novel RNA classes.

RNA structure prediction remains a challenging problem, particularly when given a single sequence and no homology structures are known, as would be the case with a genome wide search for novel RNA structures in a single species. Comparative approaches using multiple homologous sequences provide greater predictive power for RNA structure inference[25].

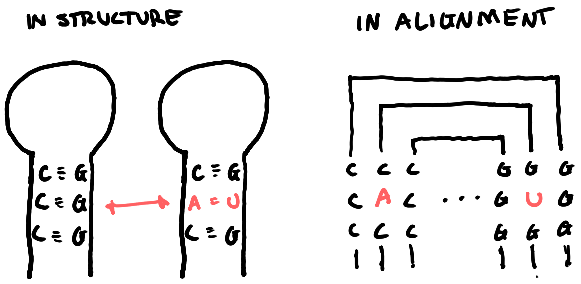


Figure 1.1: Identification of covarying sites in RNA alignments. Covarying sites are pairs of nucleotides that mutate together across evolution to preserve base pairing, providing evidence of conserved RNA structure. In this example, a C-G base pair in one species is replaced by an A-U base pair in another species, maintaining the overall secondary structure of the RNA. Covariation analysis is a powerful tool for identifying conserved RNA structures across related species, enabling the discovery of conserved structural features.

Comparative genomics leverages evolutionary conservation of RNA structures across related species, enabling the identification of conserved structural features. These conserved features are termed RNA motifs, regions showing evidence of selective constraint on structure, indicating potential functional importance. Such motifs may represent riboswitches, ribozymes, or other classes of functional RNAs.

Zasha Weinberg's lab has had notable successes with their tool CMfinder, Yao et al.[26], which identifies conserved RNA secondary structures using a comparative genomics approach. The Sack et al.[27] paper, also from the Weinberg lab, recently identified 16 new structured RNA classes in plants using a CMfinder-based bioinformatic screen. This study focused on structured RNAs involved in alternative splicing.

Roth et al.[7] also employed CMfinder to identify a new class of self cleaving ribozymes, the twister ribozyme, which has since been found to be widespread in nature. In that paper hammerhead and twister ribozymes were observed to commonly reside near each other and to certain protein coding genes, although noting it still a biological mystery. Weinberg et al.[8] followed up on this observation with a screening of sequences nearby known ribozyme and ribozyme associated protein domain coding regions. This approach successfully identified additional ribozyme classes including pistol and hatchet.

Alternate Approaches

Transcriptome mapping approaches have proven successful for unbiased riboswitch discovery. Term-seq maps RNA 3' termini genome-wide to identify premature tran-

scription termination sites characteristic of riboswitches and attenuators[28]. This approach discovered numerous antibiotic-responsive riboswitches controlling resistance genes in bacterial pathogens and the human microbiome. This approach has uncovered numerous antibiotic-responsive riboswitches controlling resistance genes in bacterial pathogens and within the human microbiome. RNA structure probing methods characterize secondary structures, tertiary interactions, and dynamic conformational changes. Selective 2'-Hydroxyl Acylation analysed by Primer Extension Sequencing (SHAPE-seq) chemically modifies unpaired nucleotides, enabling genome-wide structure mapping[29]. This approach can identify structured regions throughout transcriptomes, revealing RNA switches, ribozymes, and protein-binding sites. Dimethyl Sulfate Mutational Profiling with Sequencing (DMS-MaPseq) extends this approach to structural probing *in vivo*[30]. A variety of methods exist for studying protein-RNA interactions from both protein- and RNA-centric approaches[31].

While powerful for validation and mechanistic characterization, experimental approaches are labour-intensive, may miss RNAs with unknown functions, and are difficult to scale across many organisms.

Systematic Evolution of Ligands by Exponential Enrichment (SELEX) generates novel functional RNAs by applying selective pressure to a library of random sequences. This technique was used in the generation of the photoriboswitch, Werewolf-1[13]. Generation of synthetic functional RNAs by computational design has also been reported, with tools like RfamGen and RhoDesign[32].

Many RNA discoveries, including CRISPR-Cas systems, mRNA vaccine strategies and many catalytic RNAs were however discovered in nature, not designed *de novo*[6;16]. Beyond biotechnology, fundamental understanding of natural RNA-based systems has value, for example the discovery of flouride sensing riboswitches in bacteria

1.7 A Protein-First Approach to RNA Discovery

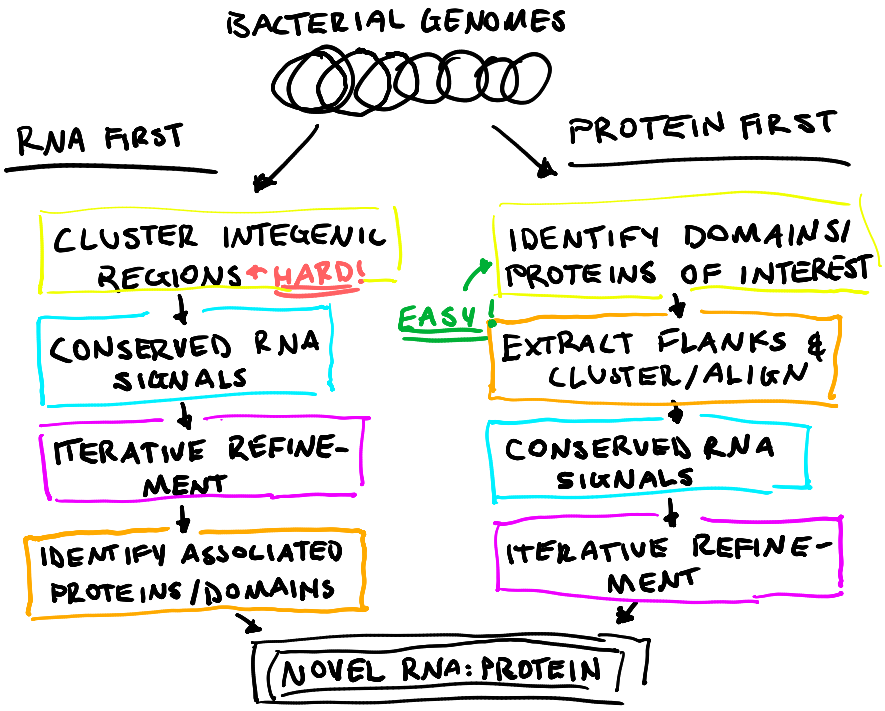


Figure 1.2: A protein-first approach to RNA discovery. Typical RNA discovery approaches are RNA-first, searching for conserved RNA structures across genomes and then associating them with nearby genes. In contrast, a protein-first approach leverages the relative ease of protein homology detection to identify conserved protein domains and deeper evolutionary relationships and then searches for conserved RNA structures in neighbouring non-coding regions.

Beyond the successes of gene family-targeted searches with CMfinder, there is precedent for riboswitches being encoded upstream of certain gene families[6;18]. This approach also provides a less computationally intensive alternative to genome-wide screens, allowing for wider searches across more species. Harnessing the power of evolutionary conservation and leveraging the biological assumption that functionally related genes (i.e. with domain homology) may share regulatory mechanisms.

Due to its potential importance in biotechnology this project will be screening for conserved RNA structures located near light-sensing gene families. Light-sensing systems in bacteria often involve complex regulatory networks, and RNA-based mechanisms have been shown to play roles in light acclimation responses in cyanobacteria[33].

A recent study by a group in Copenhagen, Geissler et al.[34], describes a bioinformatic screen similar to the one undertaken here. In that work orthologous cyanobacteria genes involved in photosynthetic pathways were targeted for nearby conserved RNA structure discovery using CMfinder. Approximately 400 novel conserved RNA

structures were identified.

Whats being improved here???

1.8 Bacterial Genomes as a Search Space

Riboswitches have been most extensively discovered and experimentally validated in bacteria and seem to rarely have homologs in eukaryotes[18]. This makes bacteria a reasonable domain for riboswitch discovery efforts.

Make a point about other functional RNAs being more widespread in bacteria, specifically RNA:protein interactions.

Extensive phylogenetic diversity is available among sequenced bacterial genomes. As of April 2026, over 870,000 bacterial genomes spanning 180,000 bacterial species clusters have been sequenced and organised in databases such as Genome Taxonomy Database (GTDB)[35]. This represents vast evolutionary distances, bacteria diverged over 3 billion years ago, providing the deep phylogenies necessary for detecting conserved structures through covariation analysis.

Bacterial regulatory mechanisms are generally simpler than eukaryotic systems, with less complex post-transcriptional processing and more direct coupling between transcription and translation.

MAYbe INCLUDE FIGURE SHOWING INCREASING NUMBER OF BACTERIAL GENOMES SEQUENCED OVER TIME, AND THE INCREASING NUMBER OF SPECIES REPRESENTED. NOW IS A GOOD TIME TO BE DOING THIS

This simplifies both computational predictions and experimental validation.

1.9 Bioinformatic Approach

Walk through the bioinformatic approach taken here, the databases used, the tools used, and the rationale for each step.

1.10 Blue Light Using Flavin Domain

BLUF

ligh switch!!!

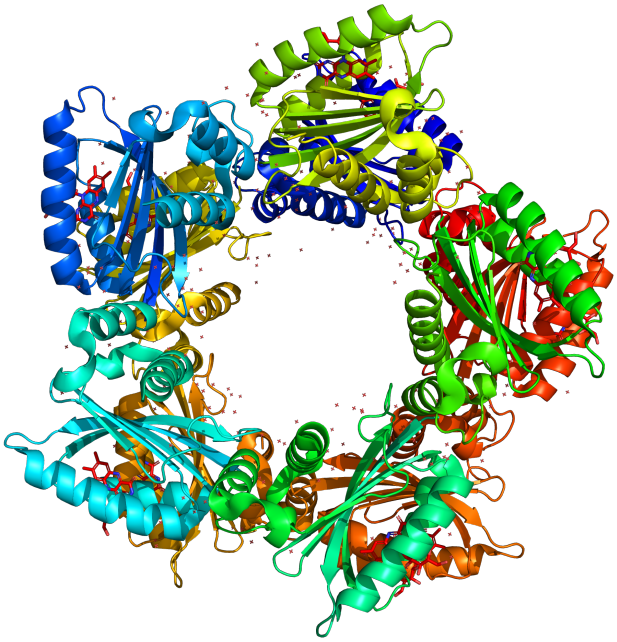


Figure 1.3: BLUF structure.

1.11 Project Aims and Hypothesis

The overarching aim is to discover novel functional RNA motifs associated with bacterial light-sensing systems by leveraging comparative genomics. Gene-family-targeted approaches offer a computationally tractable approach for identifying regulatory elements linked to specific biological pathways. Light-sensing gene families are of particular interest due to their importance in environmental adaptation and their potential relevance to biotechnological applications. Identifying conserved RNA structures associated with these genes may reveal previously uncharacterized regulatory mechanisms.

The first aim of this project is therefore to identify evolutionarily conserved RNA structural elements located near light-sensing gene families across diverse bacterial lineages. By examining intergenic and transcript regions associated with these genes, the project aims to predict conserved RNA secondary structures using comparative genomics approaches that identify covariation patterns indicative of selective constraint. These may represent riboswitches, ribozymes, or other regulatory elements.

The second aim is to develop a generalizable comparative genomics pipeline using modern tools, capable of systematically screening for conserved RNA structures linked to specific protein domain families. This pipeline is intended to be reusable and adaptable for future studies, targeting additional gene families or biological pathways.

Together, these aims will contribute both new candidate RNA motifs and a methodological platform for search for structured RNAs in nature.

Further this work aims to fill a gap in the field for a reliable statistical method for measuring relatedness between RNA elements, which is important for understanding the evolutionary relationships and structural similarities among RNA motifs.

The central hypothesis of this work is that protein homology can be leveraged in the discovery of RNA homology in RNP systems.

1.12 Vision Mātauranga

This project uses publicly available genomic datasets. If genomes from Aotearoa-associated or taonga species are included, appropriate consideration will be taken regarding data sovereignty and potential social and cultural outcomes. The Ngāi Tahu Research Consultation Committee has been contacted for consultation regarding the larger, overarching project this honors project falls under.

The research in this project aligns most closely with the Vision Mātauranga theme of indigenous innovation, contributing computational tools with potential applications in biotechnology and plant science in Aotearoa New Zealand. If this research leads to commercial outcomes, benifits of the research will be shared equitably. We remain open to collaboration with tanga te whenua.

Chapter 2

Methods

An overview of the approach used in this study is provided in Figure 2.1. The main computational discovery pipeline uses data curated from public datasets to search for conserved structured RNA elements encoded near a conserved protein domain anchor. Conversely, the positive control pipeline searches for conserved protein domains starting from a known RNA element anchor, acting to validate both the discovery pipeline and the premise of this project.

In addition to the main discovery and positive control pipelines, iterative expansion and covariance model clustering approaches are described as downstream analyses for identified candidates.

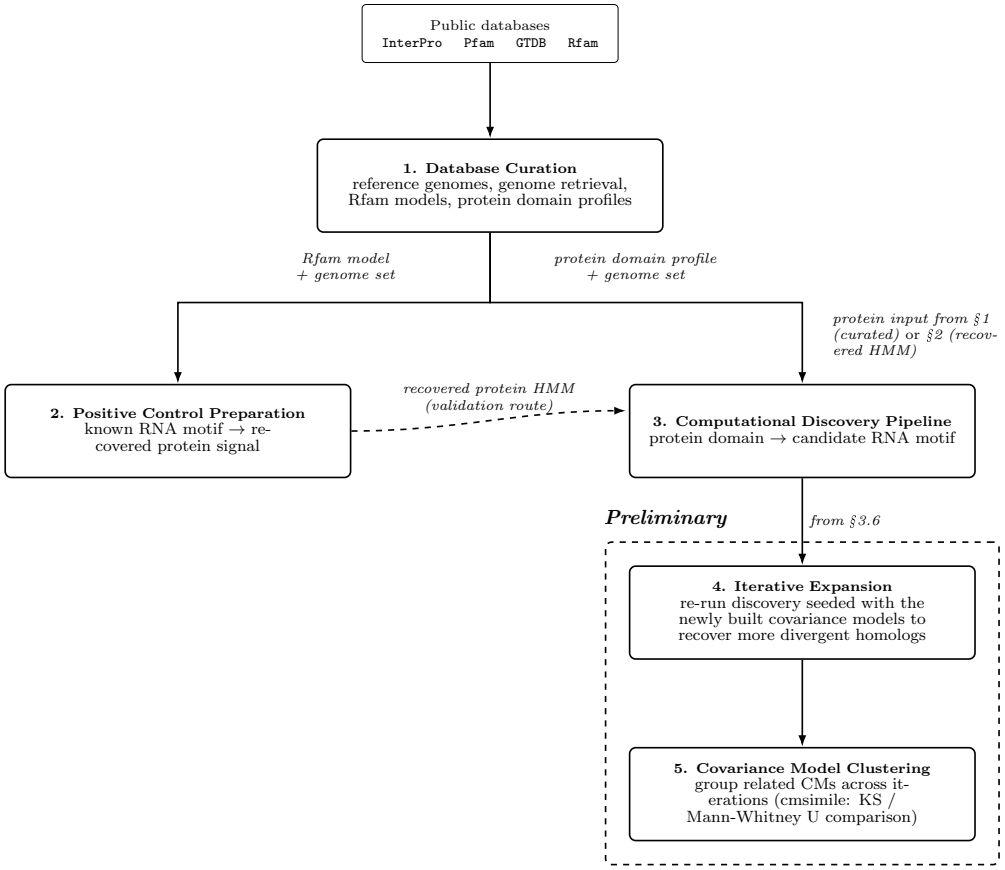


Figure 2.1: Overview of methods Simplified overview of the approach taken for RNP candidate discovery. Numbered steps refer to the following subsections.

The discovery and positive control pipelines are written in Python 3.14.6 using Snakemake v.9.23.1, and are configurable and automated[36]. Iterative expansion and covariance model clustering were developed in Go v.1.27.1 and are available as standalone tools. All code, software environment definitions and data used in this study are available in the supplementary materials (Appendix 5.1).

2.1 Database curation

2.1.1 Reference genomes

For all analyses, the GTDB release 232[37] was used to select a diverse set of bacterial genomes for analysis. The GTDB provides a standardised taxonomy, ensuring that the selected genomes are representative of the bacterial domain. Python package ete3 v.3.1.3[38] was used for phylogenetic analysis and filtering using the GTDB phylogenetic tree. Further filtering was performed to select high-quality, complete genomes (com-

pleteness $> 99.9\%$, contamination $< 1\%$), ensuring the selected reference genomes are suitable for downstream analysis. Completeness and contamination metrics were provided by CheckM2[39], available in the GTDB release.

Genome retrieval

Only genomes with genomic sequence, protein sequence and annotation files available were retrieved from the National Center for Biotechnology Information (NCBI) GenBank database release 265.0[40].

Full lists of genomes used in this study for particular experiments are available in the supplementary materials (Appendix 5.1).

2.1.2 Covariance models

RNA family covariance models were retrieved from the Rfam database (version 15.0)[41]. These were used in positive control experiments to validate the pipeline and to identify known RNA families among the candidates identified.

Additionally, covariance models and clan information available from the Rfam database (version 15.0)[41] were used to validate the clustering method developed in this study. Seed sequences (curated members of each family, used to build the alignments from which the family models are built) were also retrieved.

2.1.3 Protein domain profiles

InterPro (version 35.0) entries guided the choice of protein domain model and relevant genome search space[42]. Protein domain HMM profiles were retrieved from the Pfam v.35.0 database[43]. These were used in the discovery pipeline to search for protein homology to anchor the search for conserved structured RNA elements.

2.2 Positive control preparation

Pipeline inputs are Rfam covariance models and annotated seed sequences, as described in Section 2.1.2. A custom script accessing Entrez via the Python Biopython v.1.87 library searches for genomes containing these sequences[40]. A genome set is then constructed from GTDB reference genomes and downloaded from GenBank (Section 2.1.1). All inputs from runs of this pipeline used in this study are available in Table 5.1.

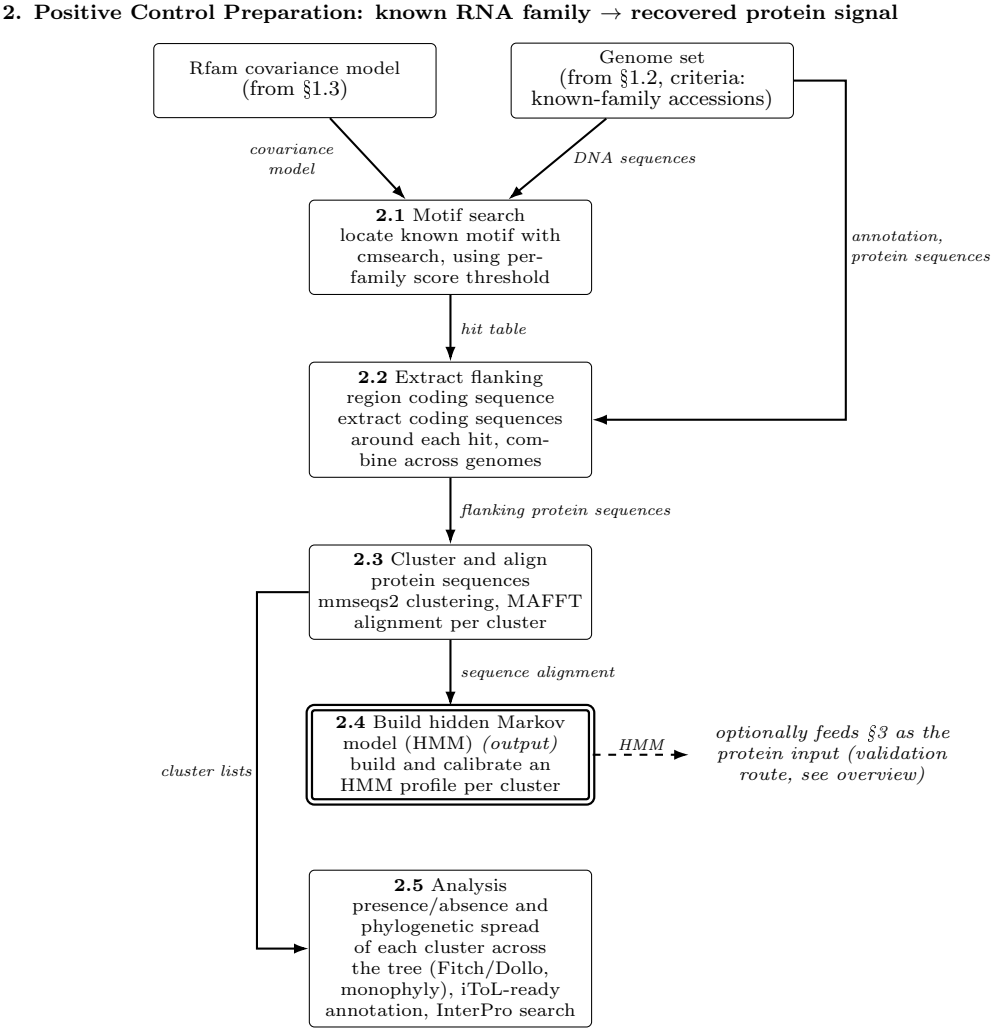


Figure 2.2: Positive control preparation (known RNA family → recovered protein signal). Starting from Rfam covariance models (§2.1.2) and genomes containing known family members (§2.1.1), the pipeline locates each known element with cmsearch, extracts the flanking coding sequences, and clusters and aligns the encoded proteins. Coding sequences within 1000 nt of each hit were extracted. Proteins were clustered with MMseqs2 (sensitivity 7.5, minimum sequence identity 0.25, coverage $\geq$ 0.2), and clusters with fewer than 5 members were discarded. Italic labels show the data passed between steps. A hidden Markov model is built and calibrated for each cluster (double border, output) and can optionally feed the discovery pipeline as the protein input. Clusters are also analysed for presence/absence and phylogenetic spread across the tree, and searched against InterPro. Hits were called using the Rfam per-family score thresholds and covariance models.

2.2.1 RNA structure search

Members of the Rfam family were searched using the Rfam published search parameters, to revalidate hits and confirm coordinate system in reference genomes (Infernal v.1.1.5 *cmsearch*). The output of this step is a list of genomic coordinates for each hit, which is used to extract the flanking coding sequences.

2.2.2 Extract flanking coding sequences

Coding sequences surrounding the RNA element locus are extracted from the genome, within a 1000 nucleotide region. This was done to identify protein domains that are encoded in close proximity to the RNA element, which may indicate a functional relationship between the two. The approach is illustrated in Figure 2.3.

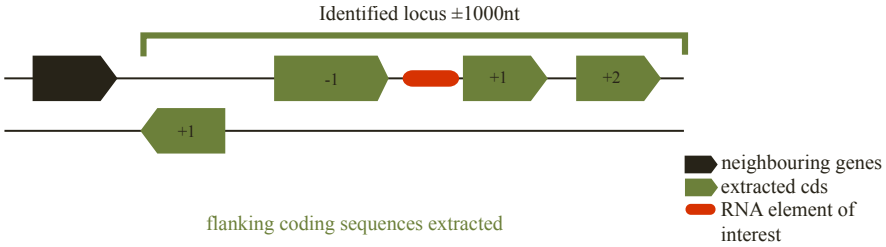


Figure 2.3: Flanking coding sequence extraction. Annotated coding sequences surrounding the RNA element of interest are extracted from genomes within a 1000 nucleotide region. Shown in red is the RNA element of interest identified by cmsearch, in black is a coding sequence outside the 1000 nucleotide extraction region. Indicated in green are the flanking coding sequences extracted for further analysis, labelled by upstream (+) or downstream (-) of the RNA element as well as the order (1 or 2 upstream or downstream). Strandedness of the surrounding coding sequences is also recorded (same or different to the cmsearch hit).

Reference genome annotation files were used to identify the coding sequences flanking the RNA element of interest, using coordinates from the RNA structure search output by a custom script. Protein sequences of these identified coding sequences were then extracted from the genome and used for downstream clustering and alignment.

2.2.3 Clustering and aligning protein sequences

Clustering

Protein homology was assessed by clustering the extracted protein sequences. Protein sequences were clustered using MMseqs2 v.14.7.0[?]. The clustering was performed with a sensitivity of 7.5, a minimum sequence identity of 0.25, and a minimum coverage of 0.2. Clusters with fewer than 5 members were discarded to ensure that only clusters with sufficient representation were retained for further analysis.

Sequence alignment

A Multiple Sequence Alignment (MSA) was built using from the trimmed cluster sequences (MAFFT v.7.525)[44]. This identifies conserved sequence features and provides a basis for secondary structure prediction and covariance model building. Alignments also describe how sequences are related to each other, which is important for downstream analyses such as covariance model building and iterative expansion.

2.2.4 Build HMM

Probabalistic models, HMMs, were built based on the MSA of each alignment. This was done to capture the sequence diversity of each cluster, and to allow for sensitive searches for homologous sequences in the discovery pipeline. HMMER v.3.4 *hmmbuild* and *hmmcalibrate* were used to build and calibrate the HMMs.

2.2.5 Analysis

The pipeline reports the results of the computational analysis, the prevelence of cluster members across the reference genomes. The position of the protein domain relative to the RNA element is also reported, as well as the number of clusters identified and their sizes.

Taxonomic spread analyses with interactive Tree Of Life (iTOL)

Monophylicity or polyphylicity of clusters was assessed by mapping the presence or absence of cluster members to the GTDB phylogenetic tree.

An annotation file was generated for each cluster, indicating the presence or absence of a cluster member in each genome. This was visualised on the iTOL v.7 online tool[45] to assess the taxonomic spread of each cluster, and identify bacterial clades where the cluster is more abundant for positive control inputs of the discovery pipeline.

Identification with InterPro

Cluster sequences were submitted to the online InterProScan v.6 tool; the results indicated the protein hits and what the protein signal represents[46]. Pfam v.35.0 entries were reported for the highest prevalence cluster, and the InterPro entry was used to identify the protein domain represented by the cluster.

2.3 Computational discovery pipeline

Inputs of the pipeline are a protein domain HMM which serves as an anchor for the search, sourced as described in Section 2.1.3, and a selection of reference genomes as a search space (Section 2.1.1). For validation the protein domain HMM is one recovered from the positive control pipeline (Section 2.2). All inputs from runs of this pipeline used in this study are available in Table 5.3.

2.3.1 Protein domain search

Protein homology was searched using a probabilistic model (HMM) against the genome sequence set (Section 2.1.1) (*nhmmer* hmmer v3.4). To narrow the search for structured RNAs, the search was focused to regions flanking the identified protein domains.

2.3.2 Flanking region extraction

The flanking regions are extracted from the reference genomes' genomic sequence based on the coordinates of the protein domain hits and the genomic annotations. The extracted sequences are then filtered for length and quality to ensure that only high-quality sequences are used in subsequent steps. Figure 2.5 illustrates the flanking region extraction process, where the anchor coding sequence is identified by *nhmmer*, and the annotated up and downstream coding sequences of the same strand are used to define the flanking regions.

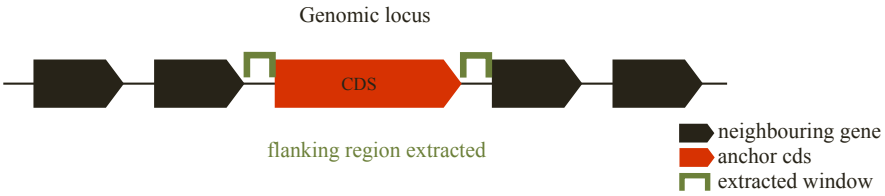


Figure 2.5: Flanking region extraction. Non-coding flanking DNA sequence is extracted from genomes based on hmmer hits in protein sequence and published genome annotations. Shown in red is the anchor coding sequence identified by nhmmer, in black are annotated up and downstream coding sequences of the same strand. Indicated in green are the flanking regions extracted for further analysis.

Length filtering included only sequences ≥ 20 nucleotides to allow for meaningful homology searches in subsequent clustering steps.

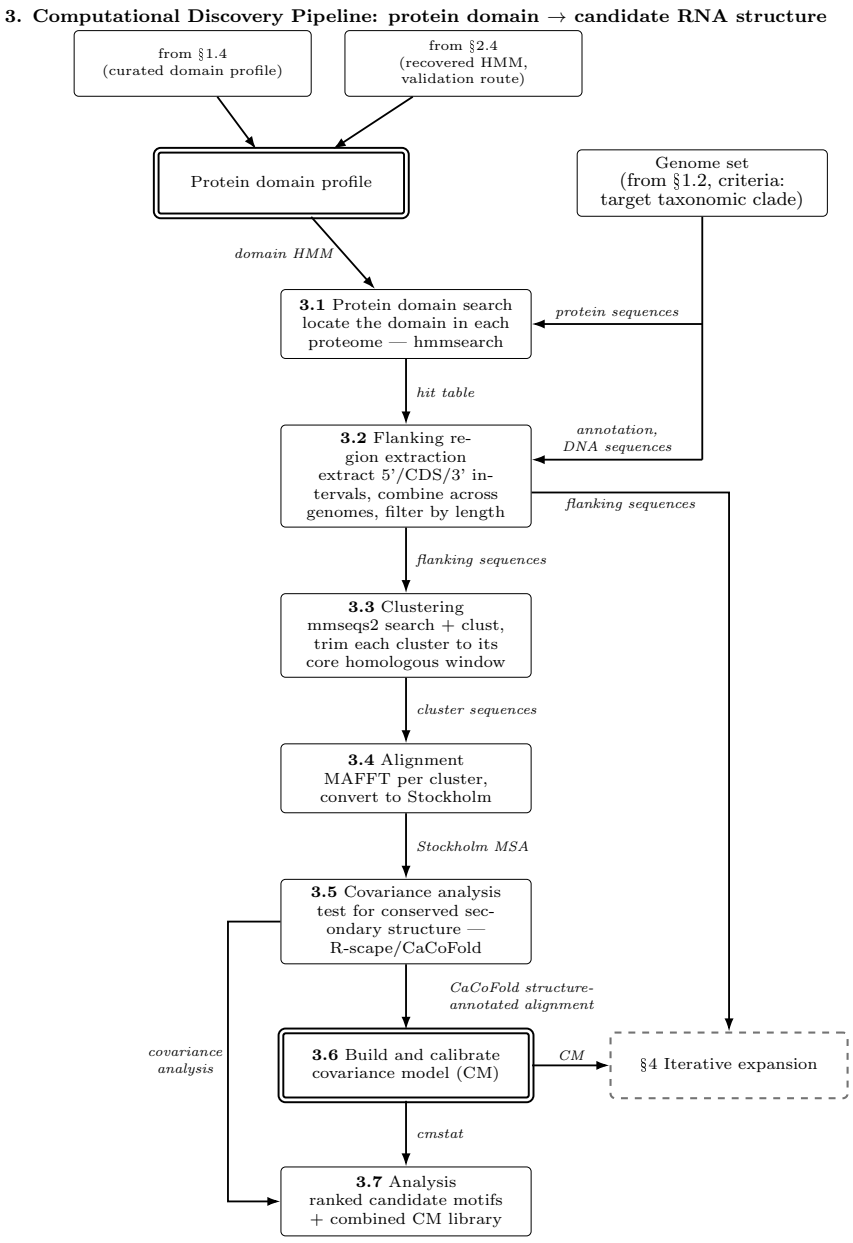


Figure 2.4: Computational discovery pipeline (protein domain → candidate structured RNA). A protein domain profile, either curated (§2.1.3) or recovered from the positive control (§2.2.4), is searched against the proteomes of the target clade (§2.1.1). Flanking regions of each hit are extracted, clustered, aligned and tested for conserved secondary structure, and covariance models are built from the structure-annotated alignments (double border, output). Candidate structured RNAs are ranked in the final report, and the covariance models and flanking sequences pass to iterative expansion (§2.4). Italic labels show the data passed between steps. Filtering thresholds are given in Table 5.2.

2.3.3 Sequence clustering

MMseqs2 clustering

Flanking sequences were clustered with a pairwise search based on sequence similarity with MMseqs2 v.16.747c6 *search*[47]. This was done to group sequences that are likely to be homologous, and to reduce the number of sequences for downstream analyses.

Sequence trimming

Clustered sequences are then trimmed to the local-alignment window it shares (directly or transitively) with an anchor sequence from the same cluster, using coordinates from a single all-vs-all search pass (MMseqs2 v.16.747c6). Window lengths were filtered to include those of length ≥ 15 and ≤ 1000 nucleotides. A padding of 15 nucleotides was kept on either side of this window. These trimming and filtering steps were implemented to remove spurious flanking sequence and to focus on the conserved region of the cluster.

2.3.4 Sequence alignment

A MSA was built using from the trimmed cluster sequences (MAFFT v.7.525)[44]. This identifies conserved sequence features and provides a basis for secondary structure prediction and covariance model building. Alignments also describe how sequences are related to each other, which is important for downstream analyses such as covariance model building and iterative expansion.

2.3.5 Covariance analysis

The CaCoFold algorithm of R-scape v.2.0.4 was used to predict conserved secondary structure and covarying base pairs in the MSA[48]. This approach tests for covariation in the alignment, and predicts a consensus structure based on the covarying base pairs. It provides a structure-annotated MSA in Stockholm format, which can be used to build a covariance model. Additionally, it provides a statistical test for covariation, giving an E-value for each predicted covarying base pair, statistical power for the covariation test, and an F-value for the overall alignment (the harmonic mean of sensitivity (SEN) and positive predictive value (PPV)). Secondary structure diagrams were generated with R2R v.1.0.6[49].

This covariance analysis step is the key step in the pipeline to identify any structured RNA elements. E-values for covarying base pairs are used to filter out spurious

predictions, and the overall F-value is used to assess the quality of the alignment.

2.3.6 Covariance model construction and calibration

Filtering was done on the CaCoFold structurally annotated alignments to include only those with at least one covarying pair where $E \leq 0.05$. The Infernal v.1.1.5 *cmbuild* tool was used to generate probabilistic models from identified CaCoFold generated structural alignments, default parameters were used. Infernal v.1.1.5 *cmcalibrate* assigns E-values to built models by fitting exponential tails to the hits of random sequences, this is required for statistical significance in searches using these models[50]. Covariance Models (CMs)s with calibration runtimes < 3 hours and requiring < 8 Gigabyte (Gb) space on 4 Central Processing Unit (CPU) cores were calibrated with `--tailp 0.05`, fitting the null tail from the top 5% of the histogram. The random sequence set size was also limited in size to 0.8 Megabyte (Mb), this trades off E-value accuracy for quicker run times.

2.3.7 Analysis

Most analysis of the discovery pipeline is done by validation with the positive control pipeline, and by downstream by iterative expansion of the covariance models (Section 2.4). Addititionally negative control protein domains were used to assess the behaviour of the discovery pipeline when applied to protein domains that are unlikely to be associated with structured RNA elements. The parameters of the clustering step were adjusted with the positive control run to find a balance between sensitivity and specificity, and to cluster size (Appendix Figure 5.1).

2.4 Iterative expansion

The idea of this iterative expansion approach is to use the probabilistic models (covariance models) built by the discovery pipeline to search for further homologous sequences in the flanking sequence set. This was done by an iterative approach described in Figure 2.6 using the Infernal v.1.1.5 toolkit[50]. The probabilistic models describe sequence and structural information, and as such can be used to search for likely homologs. These models can be used to align those homologs, and new models can be built from the resulting alignments.

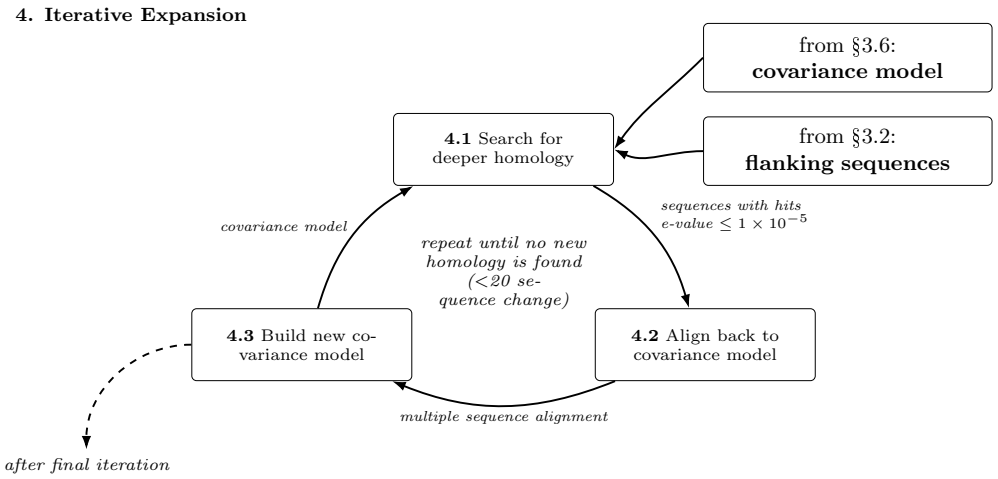


Figure 2.6: Iterative expansion of candidate covariance models. Each covariance model from §2.3.6 is used to search the flanking sequences from §2.3.2 for deeper homology (4.1) with Infernal v1.1.5. Sequences with hits at an E-value $\leq 1 \times 10^{-5}$ are aligned back to the covariance model (4.2), and a new covariance model is built from the resulting multiple sequence alignment (4.3). The cycle repeats until <20 new homologous sequences are found. The dashed arrow marks exit from the loop after the final iteration.

For all uses of this iterative expansion pipeline, a convergence cut-off of 20 new sequences identified in a round was used. This was to avoid long or infinite iterations. When this convergence limit was met a final covariance model and alignment was returned as output.

2.4.1 Search for deeper homology

The probabilistic models (covariance models) built by the discovery pipeline were used to search for further homologous sequences in the flanking sequence set (Infernal v.1.1.5 *cmsearch*). An E-value threshold of 1×10^{-5} was applied to retain significant hits, with otherwise default parameters. Resulting identified sequences were aligned back to the covariance model.

2.4.2 Align back to covariance model

Infernal v.1.1.5 *cmalign* was used to align identified homologous sequences back to the covariance model of interest[50]. This produces a structure-annotated MSA of the identified sequences, which can be used to build a new covariance model, potentally adding new structural and sequence information. These alignments were generated with default parameters as structure-annotated, Stockholm-formatted MSAs.

2.4.3 Build new covariance model

Probabilistic models describing the structure and sequence of the RNA homology identified in the previous steps were built. Infernal v.1.1.5 *cmbuild* and *cmcalibrate* were used to build and calibrate new models from the expanded MSAs[50]. Structure was preserved between rounds with the *cmbuild* `--hand` option, and default *cmcalibrate* parameters.

2.4.4 Analysis

The iterative expansion pipeline outputs are structure-annotated, Stockholm-formatted MSAss and calibrated covariance models. Reported also are the number of rounds of expansion, and the number of sequences identified in each round. Analysis of the output covariance models was done with the covariance model clustering pipeline (Section **??**). MSAs resulting from positive control candidates of the discovery pipeline were also analysed for hit identity and coverage. Hit identity was calculated as the number of identical sequences between alignments. Coverage identity was calculated using the coordinates of the alignment sequences, as the number of identical nucleotides between alignments divided by the total length of all alignment sequences. This was calculated as the intersection / union of the alignment coordinates. All analysis was done with custom R v.4.6.1 scripts, and plotted with ggplot2, available in the supplementary materials (Appendix 5.1)[51].

2.5 Covariance model clustering

An overview of the approaches taken to quantify relatedness between covariance models is provided in Figure 2.7. These approaches were developed to quantify relatedness between covariance models, and to cluster them based on this relatedness. This is described in more detail in the following subsections.

2.5.1 Sequence scoring

Sequences were scored against each model with the Infernal v.1.1.5 tool *cmsearch*, giving a log-odds bit score for each sequence against each model.

$$S_{\text{bit}}(x) = \log_2 \left[\frac{P(x \mid A)}{P(x \mid \varnothing)} \right] \tag{2.1}$$

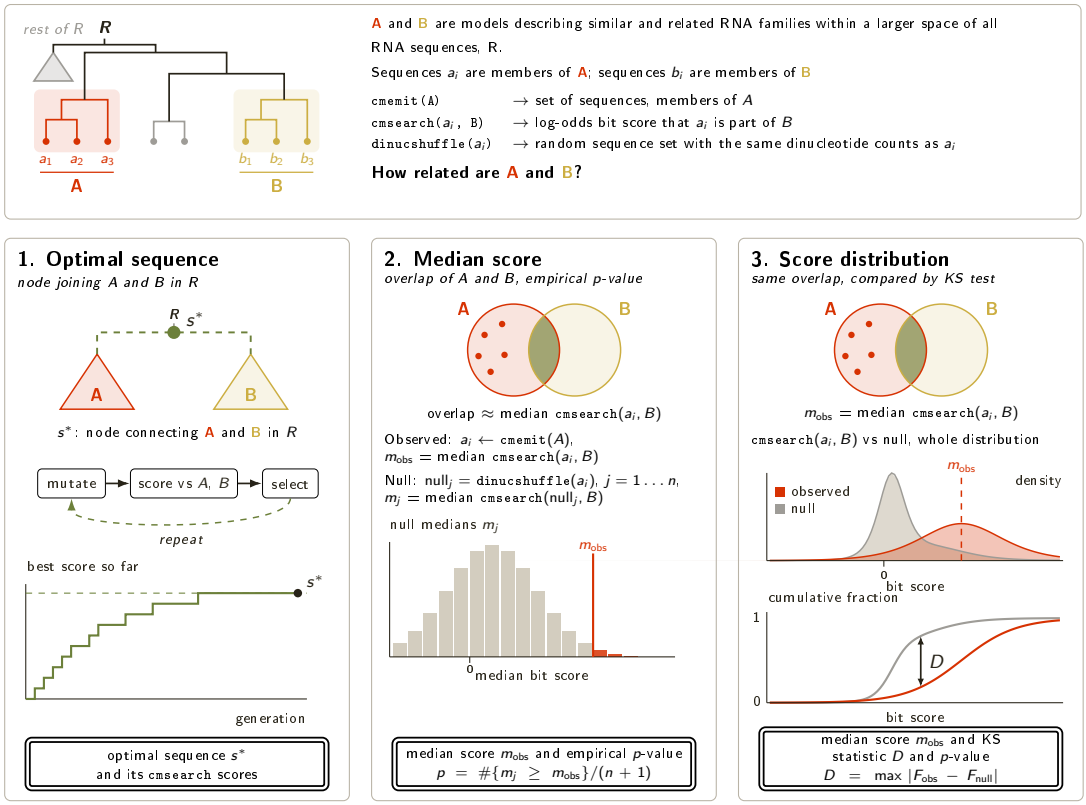


Figure 2.7: Three approaches for quantifying how related two covariance models are. Models A and B describe related RNA families within the space of all RNA sequences, R; sequences a_i are members of A and b_i are members of B. Sequences are sampled from a model with *cmemit*, scored against a model with *cmsearch* (log-odds bit score), and null sequences are generated by dinucleotide shuffling. **(1) Optimal sequence:** an evolutionary search mutates candidate sequences, scores them against both A and B, and selects the best, repeating over generations to approximate the sequence s^* that joins the two families; the scores of s^* against each model are reported. **(2) Median score:** the median bit score of sequences emitted from A and searched against B (m_{obs}) is compared with the medians of n bootstrap resamples of the pooled scores of dinucleotide-shuffled sequences (m_j), giving an empirical p-value, $p = (1 + \#\{m_j \geq m_{obs}\})/(n+1)$. **(3) Score distribution:** the full distribution of bit scores is compared with the null using a two-sample Kolmogorov–Smirnov test, where the statistic D is the maximum difference between the observed and null cumulative distributions.

where x is a sequence, A is a covariance model, and $\varnothing$ is the null model of random sequence.

2.5.2 Sequence sampling

Sequences were sampled from each probabilistic model (covariance model) with the Infernal v.1.1.5 tool *cmemit*.

2.5.3 Null sequence pool generation

Null sequences were generated by an implementation of the Dinucleotide Shuffle algorithm first described by Altschul and Erickson[52]. This approach preserves 2-mer composition while destroying any underlying structure or sequence conservation.

2.5.4 Pairwise similarity scoring approaches

Optimal sequence

This approach approximates a shared ancestor, finding a single sequence that can score highly with both models at once, based on the comparison approach by Eggenhofer et al.[53]. The optimal sequence, s^*, was approximated for each pair of models with an evolutionary algorithm. The initial population was seeded with the consensus sequence of each model and 15 sequences sampled from each model (Section 2.5.2) [, and filled to a population size of 40 by . . .]. In each of 15 generations, every sequence was scored against both models (Section 2.5.1), and the top 25% by the lower of the two bit scores were retained as survivors and parents. The population was then restored to its original size with children, each produced either by crossover of two survivors (probability 0.3) or by mutating one survivor (probability 0.7). Mutation was by point substitution: two positions chosen uniformly at random were each replaced with a nucleotide drawn at random, so sequence length was not changed by mutation. The search ran for a fixed number of generations, and the highest-scoring sequence found was taken as s^*. As the search is stochastic and limited in depth, s^* approximates the optimum and its scores are a lower bound on those of the true optimal sequence.

To make scores comparable between pairs of models, the summed bit score of s^* against both models was normalised by the scores each model gives its own sequences, to give a similarity ratio,

$$\mathrm{SR}(A, B) = \frac{s(s^*, A) + s(s^*, B)}{\bar{s}_A + \bar{s}_B} \tag{2.2}$$

where $x_1, \ldots, x_N$ are sequences emitted from model X, so that $\bar{s}_X$ is the mean self score of X. A ratio near 1 indicates that a single sequence scores about as well against both models as typical members score against their own. Implementation with a custom Go v.1.27.1 code.

Median score and empirical p-value

This approach asks how well do the sequences of one family fit into another, and vice versa. 300 sequences were sampled from each family (Section 2.5.2) and scored against the other model (Section 2.5.1).

Let $a_1, \ldots, a_N$ be sequences emitted from model A, and let $s(x, B)$ denote the *cmsearch* log-odds bit score of sequence x against model B.

$$m_{\text{obs}} = \text{median}_{i=1,\ldots,N}\ s(a_i, B) \tag{2.3}$$

To generate a null sequences the sampled sequences were dinucleotide shuffled (2.5.3). Null populations were bootstrapped from this pool with replacement and median bit scores were used to generate an empirical p-value (all with custom implementations in Go v.1.27.1).

Each a_i is dinucleotide-shuffled once, $\tilde{a}_i = \text{dinucshuffle}(a_i)$, giving a null pool of the same size as the observed set,

$$\mathcal{P} = \{\, s(\tilde{a}_i, B) : i = 1, \ldots, N \,\}\,. \tag{2.4}$$

For $j = 1, \ldots, n_{\text{boot}}$, a bootstrap population $\mathcal{P}_j^* = \{x_{j,1}^*, \ldots, x_{j,N}^*\}$ is formed by drawing N scores from $\mathcal{P}$ uniformly with replacement, with median

$$m_j = \text{median}\big(\mathcal{P}_j^*\big)\,, \qquad j = 1, \ldots, n_{\text{boot}}. \tag{2.5}$$

$$p = \frac{1 + \#\{\, j : m_j \geq m_{\text{obs}} \,\}}{n_{\text{boot}} + 1} \tag{2.6}$$

The reported metric of this approach is the sum of m_{obs} for both directions to give a single pair value.

Score distribution

The third approach taken was driven by an observation that although the magnitude of m_{obs} (Equation 2.3) is often small between same clan RNA families (Section 2.5.5),

distributional differences between $s(a_i, B)$ and $s(\tilde{a}_i, B)$ can still be seen. The two-sample Kolmogorov-Smirnov (KS) test Smirnov[54] was applied to the observed scores $s(a_i, B)$ and the null pool $\mathcal{P}$ (Equation 2.4), both of size N. The test statistic, D, is the maximum vertical distance between the empirical cumulative distribution functions of the two sets. It was computed using a custom Go v.1.27.1 implementation and is denoted D_{AintoB} for sequences emitted from A and scored against B. Reported combined score is simply $D_{AintoB} + D_{BintoA}$, used in the preliminary clan benchmark.

2.5.5 Analysis

To test the practicality and performance of the pairwise similarity scoring approaches, two complementary tests were performed: one assessed whether curated relatedness could be recovered, and the other whether fragments of a fractured RNA family could be identified as such. All curation and annotation sets and results are available (Appendix 5.1).

Data handling and plotting were done in R v.4.6.1 using the readr and ggplot2 v.4.0.3 packages[51]. Clustering was computed by pairwise distance and plotted with a dendrogram. Plotting and summary functions are available in the repository (Appendix 5.1).

Clan benchmark

Rfam families are curated into clans where they are deemed to be related and share common ancestry[41]. A set of 44 families from various clans was curated from Rfam (version 15.0) to test whether the pairwise similarity approaches could recover clan relationships. All 946 possible pairs of families were scored with each approach.

Separation of within-clan from other pairs was quantified as the area under the receiver operating characteristic curve (AUC), calculated from ranks as $\mathrm{AUC} = U/(n_{\mathrm{within}}\, n_{\mathrm{other}})$, where U is the number of (within-clan, other) pairs of pairs for which the within-clan pair has the higher statistic, counting ties as one half. This is the probability that a randomly chosen within-clan pair outscores a randomly chosen other pair. An AUC of 0.5 indicates no separation and 1 indicates perfect separation. The AUC makes no assumption about the distribution of the statistics and is unaffected by their scale. As pairs sharing a family are not independent, 95% confidence intervals were obtained by bootstrapping over families.

PROBABLY WILL REMOVE AS I DONT TALK ABOUT Fractured model

The use case of this clustering in this study was to identify candidates yielded by the discovery pipeline that describe the same signal. As such it made sense to test if the pairwise similarity scores identified similarities in covariance models describing the same underlying signal. The Rfam annotated FMN seed sequences (Section 2.1.2) were clustered, aligned, structure predicted and built into covariance models as in the final steps of the computational discovery pipeline (Sections 2.3.3, 2.3.4, 2.3.5 and 2.3.6). This set of CMs provided a test set for similarity scoring.

2.6 Ethics, MPI and EPA regulations

This project uses publicly available genomic sequence data and does not involve human participants, animal experimentation, or the construction of genetically modified organisms.

No ethical approval, MPI, or EPA approvals are required at this stage. Should the scope expand to require these approvals, they will be sought.

Chapter 3

Results

The central focus of this project was to develop a pipeline to identify structured RNA elements and to apply this in the search of candidate light-responsive RNA elements (or "light switches").

3.1 Protein homology can be rediscovered from known RNPs

The hypothesis this work is based on is that protein homology can be leveraged to identify RNA homology in RNP systems, on the basis that they are encoded at neighbouring loci. Approaching the hypothesis from the other direction, the positive control pipeline described in Figure 2.2 searches for protein homology in coding sequences neighbouring known RNA family members. Clustering of neighbouring protein sequences was used as a proxy for homology.

Table 3.1: Top cluster (by genome prevalence) for each RNA family

RNA family	Cluster	Genomes (prevalence)	Top Pfam InterProScan hit (% of cluster sequences)*	Dominant position (% sequences)**
FMN riboswitch (RF00050)	204	926/2009 (46.1%)	RibD C-terminal domain (100%)	Upstream 1 (97.9%)
SAM riboswitch (RF00162)	831	634/1535 (41.3%)	Cys/Met metabolism PLP-dependent enzyme (100%)	Upstream 1 (97.6%)
CRISPR-DR42 (RF01351)	294	28/65 (43.1%)	CRISPR-associated protein Cas2 (100%)	Downstream 1 (98.9%)
Fluoride riboswitch (RF01734)	761	259/661 (39.2%)	CrcB-like protein, camphor resistance (100%)	Upstream 1 (93.9%)

* Prevalence is the fraction of available genomes containing the cluster.
** Position is given as strand-relative direction and gene rank (e.g. upstream 1 = RNA element immediately upstream of the cluster gene).

Using protein clustering by alignment to infer protein homology in neighbouring coding sequences of RNA family members, the positive control pipeline was able to rediscover known RNP systems (Table 3.1). The FMN riboswitch, S-adenosylmethionine (SAM) riboswitch, CRISPR-DR42 and Fluoride riboswitch were all found to be associated with particular conserved protein domains, RibD C-terminal domain, Cys/Met metabolism PLP-dependent enzyme, CRISPR-associated protein Cas2 and CrcB-like protein camophor resistance, respectively. These protein domains were identified in the highest prevalence cluster of coding sequences neighbouring the RNA family members, and the position of the RNA element relative to the protein-coding gene was consistent with known biology [cite papers].

It seems likely that annotation errors play at least some role in the lack of complete consensus on position relative to homologous protein coding gene.

3.2 RNA homology can be rediscovered from known RNPs

A selection of riboswitches and the CRISPR repeat element were found to be associated with particular conserved protein domains. With this direction shown it remains to show that RNA homology can be identified anchoring the search with a conserved protein. Any of these could presumably be used for validation of the computational discovery pipeline. The CRISPR-Cas system would be interesting to validate; however, in its current state the discovery pipeline extracts only intergenic regions on the same strand as the protein anchor. Further, the prevalence of the RNP and genome availability (limited by Rfam and Entrez entries) were deciding factors. As such, the FMN riboswitch and its associated protein domain, the RibD C-terminal domain, were used for positive control validation.

3.2.1 FMN riboswitch as a positive control

The RibD C-terminal domain, with a profile HMM generated in the positive control preparation pipeline (Section 2.2.4), was used as an anchor to investigate if the FMN riboswitch signal could be rediscovered. The bacterial family Microbacteriaceae was used as a search space, informed by the phylogenetic distribution of the riboswitch and protein pairing found in the positive control preparation pipeline (Appendix Figure 5.2). 726 loci were identified in the Microbacteriaceae search space, when searched with the curated FMN Riboswitch model, E-value $< \times 10^{-5}$. The outputs of the dis-

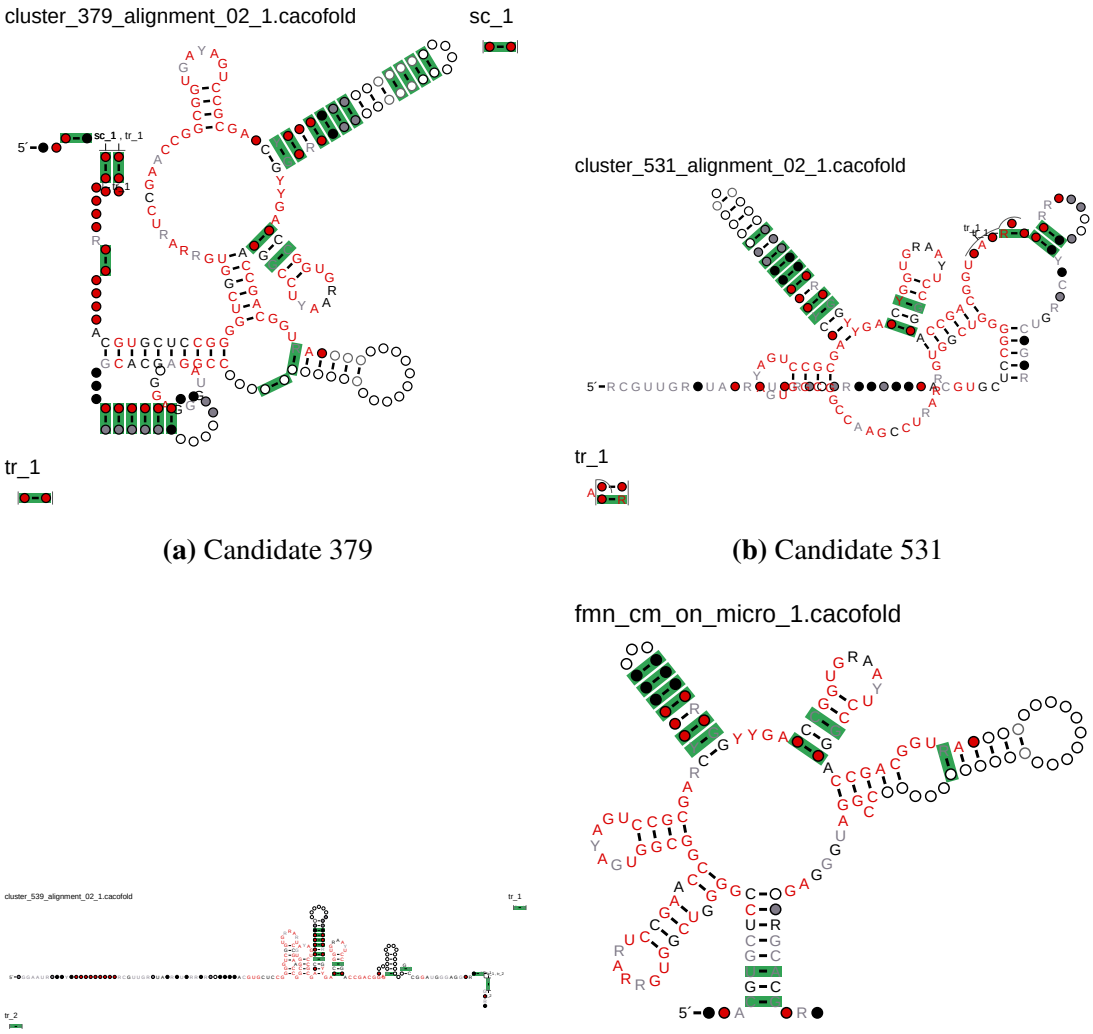
covery pipeline were run through the iterative expansion approach described in Section **??**, and the resulting alignments were analysed for covariation with R-scape[?] and compared to the true-positive alignment identified with the Rfam curated FMN Riboswitch model. Pipeline threshold for this run (and all runs) are defined in Table 5.2. The resulting candidates are shown in Table 3.2, ordered by coordinate overlap identity with the true-positive alignment, along with their covariation analysis F-scores and median summed bit scores against the true-positive alignment (both candidate into the true-positive alignment, and vice versa).

Table 3.2: FMN riboswitch rediscovery candidates. ($N = 300$ emitted sequences per model)

Candidate	F-score*	% overlap identity†	Summed median score (bits)‡
416	78.95	76.75	36.35
647	73.17	67.99	47.45
35	62.07	75.02	52.55
379	61.18	80.36	61.30
491	60.67	67.95	45.20
40	56.67	74.41	38.15
384	55.70	68.73	38.95
539	53.85	76.99	32.45
531	48.15	78.73	34.35
396	34.04	0.00	3.20
109	32.26	0.00	3.00
442	20.00	0.00	-0.60

* R-scape F-score for recovery of covarying base pairs relative to the secondary structure annotated within each candidate's own alignment, at $E \leq 0.05$.
** Fraction of the candidate's hit set (sequence accessions) also present in the true-positive alignment.
† Mean coordinate overlap (intersection/union) of the hits shared between the candidate and the true-positive alignment.
‡ Sum of the median `cmsearch` bit scores of candidate sequences scored against FMN Riboswitch and FMN Riboswitch sequences scored against the candidate.

Table 3.2 shows 9 candidates with a mean coordinate overlap of > 0.67 with the true-positive alignment, and summed median bit scores of > 32. This indicates that the candidates identify the same genomic loci as the true-positive alignment, and that the candidates' probabilistic models are describing much of the same structural information. Secondary structures were predicted for all candidates and the top 3, by highest coordinate overlap identity to the true-positive alignment are shown in Figure 3.1. The predicted structures of the remaining candidates are shown in Appendix Figure 5.4.



(a) Candidate 379

(b) Candidate 531

(c) Candidate 539

(d) Positive control

Figure 3.1: Predicted covarying secondary structures of the three FMN riboswitch candidates with highest coordinate overlap identity to the positive control. R-scape CaCoFold structures generated from iteratively expanded alignments, ranked by mean coordinate overlap (intersection/union) of hits shared with the Rfam-annotated true-positive alignment. **(a)**:– textbf(d): show the three candidates with highest overlap identity; **(d)**: shows the same analysis run on the true-positive alignment (`fmn_cm_on_micro`) for comparison.

The CaCoFold predicted structures of the three candidates with highest coordinate overlap identity to the true-positive alignment (Figure 3.1) illustrate the presence of covarying base pairs in the same regions as the true-positive alignment, and the overall secondary structure is similar (Figure 3.1).

3.2.2 Negative control validation

Four membrane-associated protein domains were chosen as negative controls on the basis that they were unlikely to be involved in RNP systems. One of the four do-

mains returned promising structured RNA candidates from the computational discovery pipeline [talk about in discussion] The search space for all was the genus *Burkholderia*, which was chosen as all four domains are present in this genus.

Table 3.3: Negative control candidates.

Label	IPR	Candidate	Direction	Cov. pairs*	F-score
SecY	IPR002208	26	Downstream	1	6.06
SecY	IPR002208	98	Downstream	1	4.44
YgaP-like	IPR021309	90	Downstream	2	5.63

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 in all cases.
† No F-score available: the covariance model built for this cluster contained zero base pairs, so no reference structure existed against which to compute recovery statistics.
TolC (IPR005594, PF03895) was also screened in both directions (3 clusters total) but produced no clusters with a significant covarying pair.

Of the three negative control domains, only the SecY and YgaP-like inner membrane protein domains returned candidates with at least one covarying pair, E-value ≤ 0.05 (Table **??**). Of these the F-scores are low, and the number of covarying pairs is small, compared to the positive control candidates, indicating that these are likely false positives, illustrating the selective nature of the discovery pipeline. These results outline expected behaviour of the discovery pipeline when applied to protein domains that are unlikely to be associated with structured RNA elements, and provide a baseline for comparison with the positive control results.

3.3 BLUF domain RNP search

Having tested the discovery pipeline with positive and negative controls, the pipeline was applied to the BLUF domain, which is a light-responsive protein domain. The BLUF domain anchored search was conducted in *Burkholderia*, as this genus contains a 701 BLUF domain-containing protein sequences[42].

3.3.1 Identified candidates

The discovery pipeline identified two promising structured RNA candidates, Candidate 130 and Candidate 332, which were found in the downstream flanking sequences of BLUF domain-containing proteins. These candidates were additionally analysed with the iterative expansion approach described in Section **??**, and the resulting alignments were analysed for covariation with R-scape[?] .

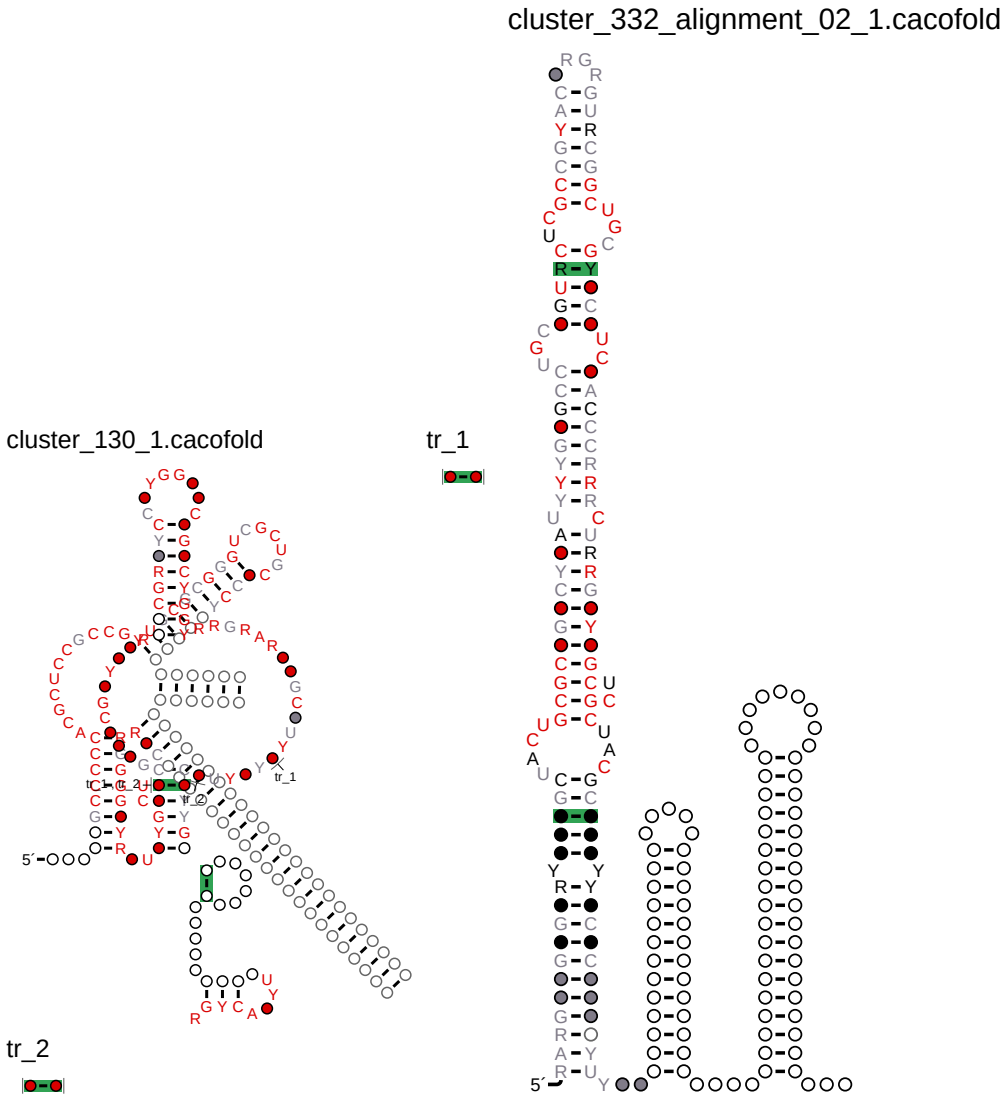
Table 3.4: Covariation analysis for the two BLUF-associated candidates, before and after iterative expansion of the covariance model.

Candidate	Stage	Cov. pairs*	F-score
130	Pre-expansion	4	11.27
130	Post-expansion	0	0.00
332	Pre-expansion	1	2.33
332	Post-expansion	2	5.00

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 in all cases.

Iterative expansion yielded more sequences for both Candidate 130 and 332, the number of covarying base pairs decreased for Candidate 130 from 4 to none, and increased for Candidate 332 from 1 to 2, similarly the F-score increased for Candidate 332 and decreased for Candidate 130 (Table 3.4). This indicates that the original alignment for Candidate 130 better describes a structured element, while the iterative expansion approach improved the alignment for Candidate 332.

With F-scores of 11.27 and 5.00, respectively, both candidates show some evidence of covariation, but similar strength to that of the negative control candidates. With 4 and 2 identified covarying base pairs for these candidates is higher than that of the negative control candidates, suggesting there may be some structural signal present. Predicted secondary structures are shown in Figure 3.2.



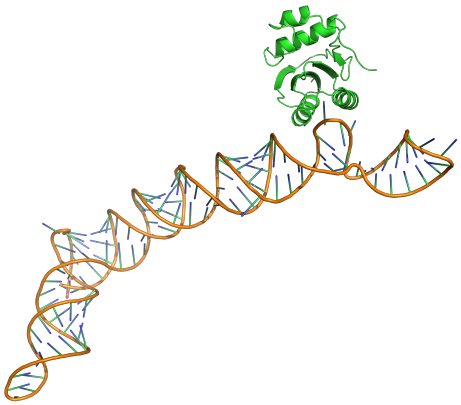
(a) Candidate 130 **(b)** Candidate 332

Figure 3.2: Predicted covarying secondary structures of the two BLUF-associated candidates. Predicted covarying secondary structures of the two most promising structured RNA elements identified through the discovery pipeline. Structures are generated by R-scape CaCoFold. **(a)**: shows the predicted secondary structure of candidate 130 as output by the computational discovery pipeline described in the methods, **(b)**: shows that for candidate 332 run through the iterative expansion approach described in the methods.

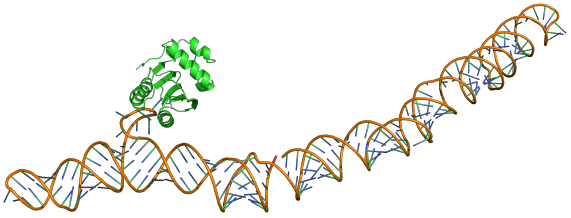
Candidate 130 shows some promise for an alternative structure with a conserved CGGCG and GCCGC sequence not predicted to base pair in the CaCoFold structure, but which could form a stem-loop structure with covarying base pairs.

3.3.2 RNP complex prediction

3D structure and interaction prediction was done with AlphaFold 3[?] using the highest-confidence model for each candidate.



(a) Candidate 130



(b) Candidate 332

Figure 3.3: Predicted ribonucleoprotein complex structure of BLUF with identified RNA candidates. Predicted structures of representative RNA and protein sequences from the two most promising structured RNA elements identified through the discovery pipeline. Predictions were made with AlphaFold 3 and the highest-confidence model is shown. **(a)** shows the predicted structure and interaction of candidate 130 with the BLUF containing protein it was found near, **(b)** shows that for candidate 332.

RNA binding is predicted around the BLUF FAD binding site identified by Kita et al.[55].

3.4 Covariance model clustering

Clustering covariance models (CMs) requires a pairwise score that is both reliable and statistically robust, so that a distance between two models can be interpreted rather than just ranked. This section describes how candidate scoring approaches were benchmarked against Rfam clans, and how the chosen approach was then applied to cluster the positive control candidates.

Rfam groups families into clans on the basis of shared evolutionary origin, which makes clan membership a useful, if imperfect, ground truth for benchmarking: models in the same clan should recognise each other's sequences, whereas models from different clans should not. Rfam clan membership was therefore used to benchmark the pairwise scoring methods.

3.4.1 Clustering approaches

Three approaches were tested: [one which searched for a common ancestor with a genetic algorithm, one which ..., and one which ...]. Each was evaluated on its ability to distinguish within-clan pairs from all other pairs (Table 3.5).

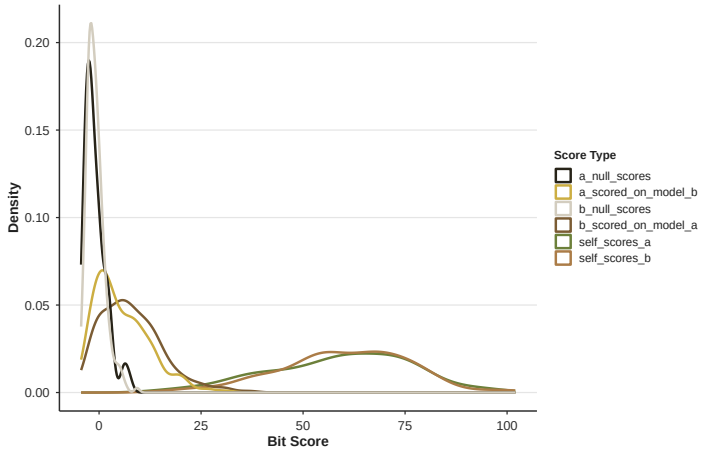
Table 3.5: Separation of within-clan from other pairs by each scoring approach. Within-clan ($n = 27$) and other ($n = 919$) pairs from annotated Rfam entries. AUC is the probability that a randomly chosen within-clan pair scores higher than a randomly chosen other pair; p-values are from a Wilcoxon rank-sum test.

Approach	Statistic	AUC	95 % CI
Common ancestor (similarity ratio)	similarity ratio	0.832	[0.716 – 0.934]
Median bit score vs. null	summed medians	0.767	[0.560 – 0.944]
KS distribution vs. null	$D_{A\to B} + D_{B\to A}$	0.835	[0.653 – 0.957]

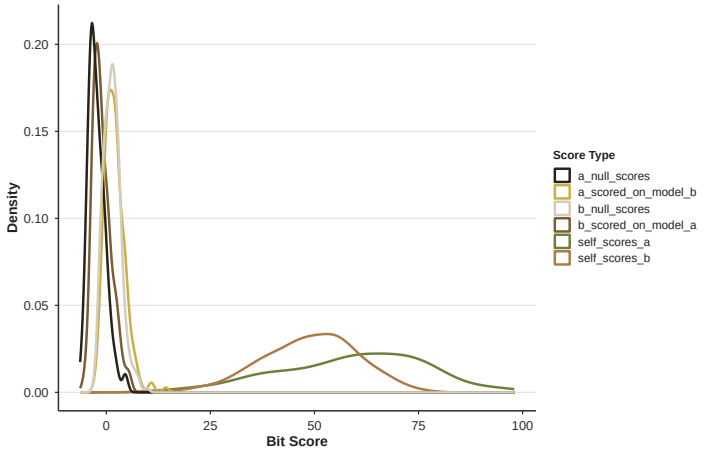
All three approaches separated within-clan pairs from other pairs. The similarity ratio and the summed KS statistic performed almost identically (AUC = 0.832 and 0.835), while the summed median bit score against the null performed less well (AUC = 0.767). Because the KS approach provides a per-pair test against a null distribution, which the similarity ratio does not, it was used for all remaining analyses.

3.4.2 Distributional differences to the null

The KS approach compares the bit scores obtained when one model scores the sequences of another against a null distribution of shuffled sequences, in each direction. Figure 3.4 shows this for two pairs.



(a) SNORA17 vs. SNORA43



(b) SNORA17 vs. Pyrobac-1

Figure 3.4: Distributional comparison of bitscores for a within-clan and a between-clan model pair. Cross comparison bit score distributions are compared to null sequence distributions. Self bit scores (sequences scored against their model of origin) are shown for comparison. **(a)**: SNORA17 (a) vs. SNORA43 (b), both members of the same clan; the cross-comparison distribution (SNORA43 sequences scored against the SNORA17 model) is compared to null sequences (KS $D = 0.673$, $p = 1.66 \times 10^{-19}$ scored against SNORA43; $D = 0.557$, $p = 3.22 \times 10^{-30}$ scored against SNORA17). **(b)**: SNORA17 (a) vs. Pyrobac-1 (b), from different clans; (KS $D = 0.263$, $p = 5.26 \times 10^{-7}$; $D = 0.112$, $p = 0.112$).

For SNORA17 and SNORA43, which belong to the same clan, the cross-comparison scores were strongly shifted from the null in both directions ($D = 0.673$ and $D = 0.557$; $p = 1.66 \times 10^{-19}$ and 3.22×10^{-30}). For SNORA17 and Pyrobac-1, which belong to different clans, the shift was much weaker in one direction ($D = 0.263$, $p = 5.26 \times 10^{-7}$) and not significant in the other ($D = 0.112$, $p = 0.112$). Summing the two directional statistics therefore gives a single, symmetric measure of mutual recognition.

Across all pairs tested, the summed KS statistic was significantly higher for within-clan pairs ($n = 27$) than for other pairs ($n = 919$; Wilcoxon $p = 2.76 \times 10^{-9}$; Figure 3.5). Because every model is compared against a single null population rather than a separate null for each pairwise comparison, this separation does not depend on pair-specific null generation, which is also considerably cheaper to compute.

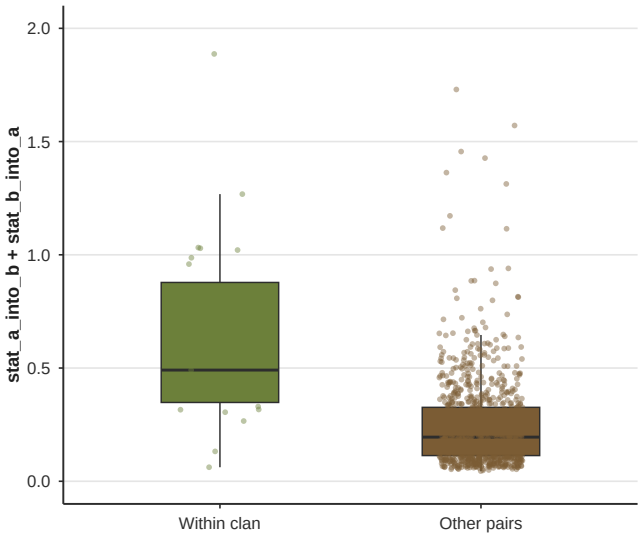
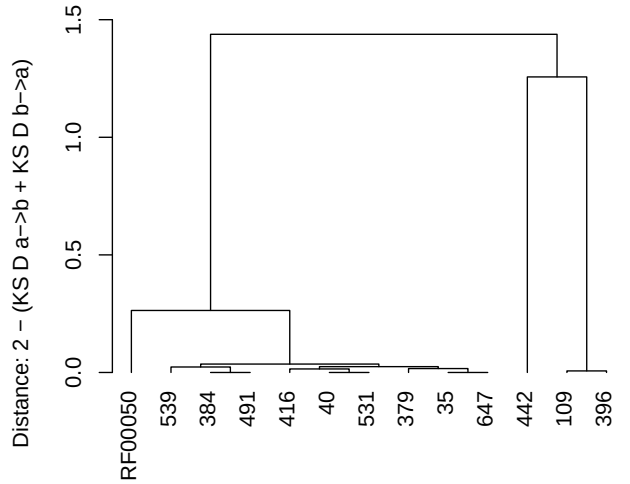


Figure 3.5: Within- vs. between-clan separation using the KS statistic compared against a single null population. Unlike the bitscore-median approach, this compares the full score distribution to only one bootstrap null population per model rather than one null per pairwise comparison, separation of the approach tested (Wilcoxon $p = 2.76 \times 10^{-9}$; Table 3.5). The y-axis is the sum of 2 two-sample KS statistics, D, of the bit score distribution against the null (both D_{AintoB} and D_{BintoA}). A pair is labelled within-clan ($n = 27$) if both families belong to the same Rfam clan, and other ($n = 919$) otherwise, including pairs involving a family with no clan assignment.

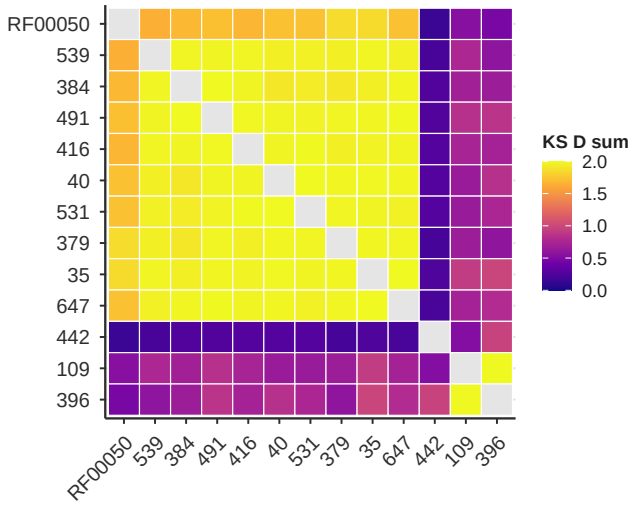
3.4.3 Clustering positive control candidates

The summed KS statistic was used to cluster the candidate models together with the FMN riboswitch model (RF00050), using the distance $2 - (D_{A \to B} + D_{B \to A})$ and average-linkage hierarchical clustering (Figure 3.6). The resulting tree shows candidates clustering into three main groups, one with the FMN riboswitch model clustering with candidates 539, 384, 491, 416, 40, 531, 379, 35 and 647, which were also the candidates with the highest coordinate overlap identity to the true-positive alignment (Table 3.2). Candiates 442, 109 and 396 do not cluster with the FMN riboswitch model, and are therefore likely to be distinct from the true-positive alignment. 109 and 396 cluster together, which can also be observed from their predicted secondary structures (Appendix Figure 5.5). In downstream analysis these candidates could be combined

into a single model, potentially improving the alignment and covariation signal.



(a)



(b)

Figure 3.6: Clustering of candidate covariance models by cross-recognition. **(a)** Hierarchical clustering (average linkage) of the candidate models and the FMN riboswitch model (RF00050), using the distance $2-(D_{A\to B}+D_{B\to A})$, where D is the two-sample Kolmogorov–Smirnov statistic against the shuffled null in each direction. **(b)** Pairwise summed KS statistic, with models ordered as in (A); higher values indicate that the two models recognise each other's sequences more strongly. Candidates are labelled by cluster number.

Chapter 4

Discussion

4.1 Discovery pipeline

4.1.1 Postive control set

FMN riboswitch chosen for its clear biological association with a protein and relative conservation across a wide set of bacterial species.

CRISPR one too

4.1.2 Negative control set

There are many dificulties in the curation of a negative control set for a discovery pipeline stemming from the difficulty of proving a negative.

What it showed

4.1.3 Iterative expansion approach

4.1.4 Searching with BLUF

BLUF was chosen as a proof-of-concept search target for its potential to be involved in an RNP 'light switch' complex.

4.2 Covariance model clustering appraoch

4.2.1 Why

There is a hole in bioinformatic tools for a fast, working, reliable and statistically defensible similarity scoring. C

4.2.2 Applications

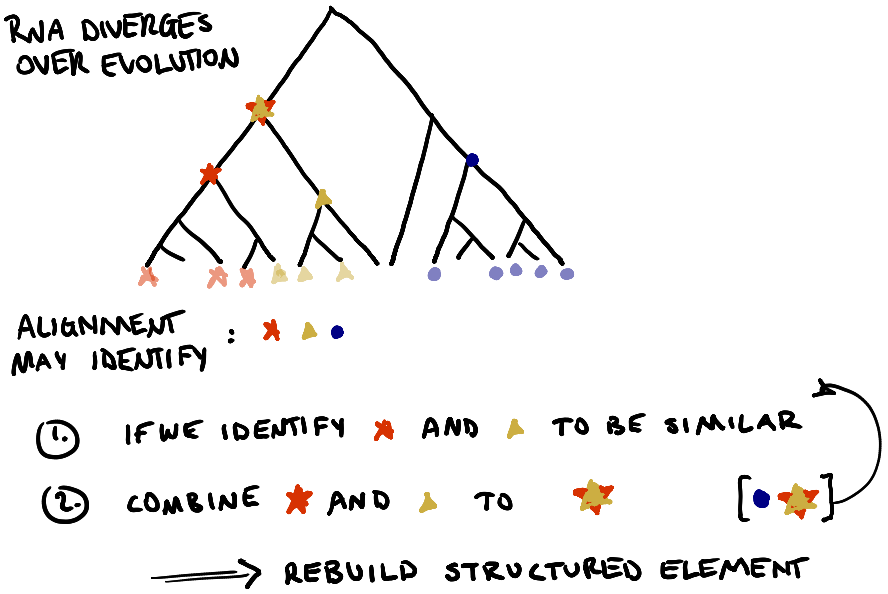


Figure 4.1: Covariance model clustering and merging. Shown in red, gold and blue are structured RNA elements that have been identified by local alignment. Identified elements are homologous to eachother, but are not similar enough to be detected by classical homology search methods. Covariance models describing these elements can be clustered based on their similarity, and merged to create a new model that captures the shared features of the original models. This approach can be used to identify more distant homologs and improve the sensitivity of RNA discovery pipelines.

4.3 Limitations

Currently looking over flanking regions which include coding regions of opposite strand - will be picking up on coding level conservation here

Limited to just up and down stream of same strand, as we saw with the CRISPR-DR42 conserved associations can go beyond this limited relationship

Relying on unreliable annotations

Should have used McGimpsey's iterative nhmmer approach for a deeper protein homology search

Only ran a single relativey easy positive control, FMN ribiswitch on a protein domain profile generated specifically on the basis of association to the Rfam annotated riboswitch model.

4.4 Future directions

4.4.1 Discovery pipeline extension

Improve tool selection, filtering steps and tool parameters fo

Structural clustering and alignment for RNA exists

Incorporate covariance model clustering and combination

Apply cluster-merging to strengthen signal to identify more distant homology.

Investigate introducing binary logic (AND, OR, NOR) to the protein domain selection

4.4.2 Further targets for RNP discovery

Expand beyond BLUF to other light-pathway anchors (e.g. phytochromes, cryptochromes). And more!

4.4.3 Experimental validation

Expand search for promising candidates and experimentally validate (chemical probing with SHAPE)

4.5 Conclusions

A protein first approach works for RNA discovery. Under limited RNP or cis-regulatory element cases blah blah...

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Chapter 5

Appendices

5.1 Code and Data Availability

Code is available from:

- https://github.com/dan-432/DarwinsRNAHunt contains the positive control and discovery snakemake pipelines and associated analyses.
- https://github.com/dan-432/cm-iterative-expansion contains the logic for and analysis of the itertive expansion of candidate CMs.
- https://github.com/dan-432/cm-simile contains the logic for and analysis of the pairwise similarity scoring of candidate CMs

See top-level README files for further documentation on file structure and running instructions.

Data and computed results are available from . . .

[link to zenodo]

5.2 Supplimentary Tables

Table 5.1: Rfam families used as positive controls for the RNA-to-protein direction.

Rfam ID	Description
RF01315	CRISPR RNA direct repeat element
RF00050	FMN riboswitch
RF00162	SAM riboswitch
RF01734	Fluoride riboswitch

Table 5.2: Filtering thresholds used in the computational discovery pipeline.

Step	Parameter	Value
3.1 Domain search	hmmsearch E-value	$\leq 10^{-5}$
3.2 Flanking extraction	Minimum length	20 nt
	Overlap into coding sequence	5 nt
3.3 Clustering	mmseqs2 minimum sequence identity	0.4
	Coverage (mode 3)	≥ 0.5
	Minimum cluster size	5
	Trimming padding	15 nt
3.5 Covariance analysis	R-scape E-value (at least one covarying pair)	≤ 0.05
3.6 Calibration	Random sequence length	0.8 megabases

Table 5.3: Inputs to the protein-to-RNA discovery pipeline: protein domain (or protein cluster), description, and the GTDB bacterial clade searched. Genomes were filtered at 99.9% completeness and $\leq$1% contamination for every pair.

Domain ID	Description	Bacterial taxon (GTDB node)	Role
IPR005594	SecY domain	*f__Burkholderiaceae_C* (36.0)	Negative control
IPR002208	YadA-like membrane anchor domain	*f__Burkholderiaceae_C* (36.0)	Negative control
IPR021309	YgaP-like transmembrane domain	*f__Burkholderiaceae_C* (36.0)	Negative control
IPR001029	Bacterial flagellin N-terminal helical region	*f__Burkholderiaceae_C* (36.0)	Initially as negative control
IPR007024	BLUF domain	*f__Burkholderiaceae_C* (36.0)	Test target
cluster_1341	FMN riboswitch control (protein cluster)	*f__Microbacteriaceae* (100)	Positive control
CRISPR_Cas2	CRISPR RNA direct repeat element control (Cas2)	*o__Lactobacillales* (92.0)	Positive control

Table 5.4: Top 10 clusters by genome spread for each RNA family.

Cluster	Genomes	Seqs	Prev.	Position	Up.	Cont.	Strand	Interg.	Purity
FMN riboswitch (RF00050): 1599 clusters, 2009 genomes, 1115 singleton-genome clusters									
204†	926	934	0.4609	Up 1	0.913	0.007	1.0	0.979	0.4609
695	578	748	0.2877	Up 1	0.999	0.003	1.0	0.979	0.288
600	409	417	0.2036	Up 1	1.0	0.01	1.0	0.981	0.3223
89	295	295	0.1468	Up 1	1.0	0.01	1.0	0.98	0.147
58	189	189	0.0941	Down 1	0.0	0.0	1.0	0.979	0.1547
340	188	188	0.0936	Up 1	1.0	0.005	1.0	0.973	0.2709
1453	119	119	0.0592	Down 2	0.0	0.0	1.0	0.975	0.1941

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Table 5.4 – continued

Cluster	Genomes	Seqs	Prev.	Position	Up.	Cont.	Strand	Interg.	Purity
100	96	99	0.0478	Down 1	0.111	0.0	0.869	1.0	0.0478
1501	90	93	0.0448	Down 1	0.462	0.0	0.72	0.935	0.0448
113	71	71	0.0353	Up 1	1.0	0.0	0.0	0.958	0.7474
SAM riboswitch (RF00162): 2247 clusters, 1535 genomes, 1553 singleton-genome clusters									
831†	634	873	0.413	Up 1	0.885	0.006	0.982	0.976	0.413
252	448	450	0.2919	Up 1	0.847	0.082	0.98	0.913	0.2919
2042	406	413	0.2645	Up 1	0.961	0.0	0.884	0.99	0.2645
1591	401	412	0.2612	Up 1	0.993	0.005	0.993	0.973	0.2612
1392	361	586	0.2352	Up 1	0.931	0.015	0.945	0.966	0.2352
2115	262	304	0.1707	Up 1	0.947	0.013	0.977	0.961	0.1707
1725	198	198	0.129	Up 1	1.0	0.0	0.0	0.864	0.3984
1834	134	134	0.0873	Up 1	1.0	0.0	0.0	1.0	0.0873
1517	116	138	0.0756	Up 1	0.992	0.058	1.0	0.935	0.0756
833	98	99	0.0638	Down 1	0.0	0.0	1.0	0.96	0.5904
CRISPR-DR42 (RF01351): 394 clusters, 65 genomes, 340 singleton-genome clusters									
294†	28	264	0.4308	Down 1	0.011	0.0	0.0	0.989	0.4308
217	27	192	0.4154	Down 2	0.068	0.0	0.0	0.99	0.4154
134	24	419	0.3692	Up 1	0.577	0.007	0.024	0.981	0.3692
162	7	65	0.1077	Down 1	0.0	0.0	0.015	1.0	0.1077
121	6	238	0.0923	Up 1	0.542	0.0	0.891	1.0	0.1364
161	6	69	0.0923	Up 1	0.565	0.0	0.188	1.0	0.1364
301	5	37	0.0769	Up 1	1.0	0.0	0.0	1.0	0.0769
120	4	34	0.0615	Down 1	0.0	0.0	0.735	1.0	0.0615
136	4	12	0.0615	Down 2	0.25	0.0	0.333	1.0	0.0615
253	4	20	0.0615	Up 1	0.85	0.0	0.7	1.0	0.0615
Fluoride riboswitch (RF01734): 951 clusters, 661 genomes, 779 singleton-genome clusters									
761†	259	345	0.3918	Up 1	0.974	0.014	0.977	0.939	0.39248
453	254	269	0.3843	Up 1	0.992	0.019	0.981	0.944	0.3848
768	73	89	0.1104	Up 2	0.909	0.011	0.91	0.899	0.1106
561	64	65	0.0968	Up 1	1.0	0.062	1.0	0.877	0.2623
136	33	43	0.0499	Up 1	0.814	0.0	0.744	0.977	0.05
11	28	39	0.0424	Up 1	0.895	0.026	0.795	0.949	0.0424
577	21	28	0.0318	Up 1	1.0	0.036	0.643	0.929	0.0318
169	19	20	0.0287	Down 1	0.4	0.0	0.75	0.85	0.0288
526	18	23	0.0272	Up 1	0.739	0.0	0.87	0.913	0.0272
279	17	17	0.0257	Up 1	0.941	0.0	0.059	0.824	0.0258

Genomes/Seqs: number of genomes/sequences in the cluster; Prev.: fraction of available genomes containing the cluster; Position: dominant position relative to the nearest gene (upstream/downstream and gene rank); Up., Cont., Strand, Interg.: fraction of sequences upstream of, contained within, on the same strand as, and fully intergenic relative to that gene; Purity: clade purity; All clusters were polyphyletic. † marks clusters with an InterProScan hit (see Table 3.1).

Table 5.5: Full R-scape covariation results and pairwise identity summary for FMN riboswitch rediscovery, and for the Rfam-annotated true-positive alignment.

Candidate	Cov. pairs*	True	FP	E-value range	% id. true pos.	% id. nearest
fmn_cm_on_micro (positive control)	11	42	0	$[-9.95, 874.13]$	100.00	80.36 (379)
416	30	46	0	$[-9.95, 916.18]$	76.75	94.87 (531)
647	30	52	0	$[-9.95, 923.72]$	67.99	96.65 (491)
35	27	60	0	$[-9.95, 903.89]$	75.02	92.87 (379)
379	26	59	0	$[-9.95, 909.40]$	80.36	92.87 (35)
491	27	62	0	$[-9.95, 917.35]$	67.95	96.65 (647)
40	17	43	0	$[-9.95, 918.20]$	74.41	92.95 (531)
384	22	57	0	$[-9.95, 934.88]$	68.73	94.06 (647)
539	14	38	0	$[-9.95, 898.55]$	76.99	87.36 (384)
531	13	41	0	$[-9.95, 910.91]$	78.73	94.87 (416)
396	8	39	0	$[-9.95, 822.31]$	0.00	90.79 (109)
109	5	26	0	$[-9.95, 829.86]$	0.00	90.79 (396)
442	2	18	0	$[-9.95, 491.32]$	0.00	9.71 (396)

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 for all alignments.

Table 5.6: My table caption

ID	Description	Sequence
cds-APW43854.1|utr3|CP019239.1:3318396-3318523(-)		
	Representative from Bluf downstream candidate 130	
	RNA	ACGUGGCCCCCACGCUCCGCCGCUGCGCGG GUCGCUGCCCCCCGGGGGGGCUAAUUUGGC UUCGCCGCGCUUCGCGAGCCCUUGGGGCGG CCCGGCGGGAAAUUGCCUUUCCUUCCUCCG UCUACUGA
	Protein	MLVRLLYASRAIDSRPDAIEAILTQSRNYNPSC GITGILCYGGGIFLQAIEGGRMAVSELYGHIQK DARHKDVVLLHYEEISERRFGGWTMGQVNMS KINTNILLKYAEKPELDPYSVSGKVSLALLEEL MATASIIGRA
cds-MFN4120999.1|utr3|JBKBKQ010000028.1:10309-10517(-)		

Continued on next page

Table 5.6 – continued

ID	Description	Sequence
	Rep from candidate cluster 332	
	RNA	AAAGCGGGGCGCUUUGCUACUGCGCCAGCC GGUUCCGGGCCUGCUGUGCUCGCCGUACAG GAGUACGGCUGCGCGCAUCGACCCAGACUA GGCUGGCUCGCUACACAAAUCGCCCCACCU UUCUUCGCAGUCAUCCAUCACGGUUCGUAG CCGCAGCCGCCUAGAAGGCGCUGCGGCUUU UCUGUUCAGGCCGUUCUUUUUCUCGCCCG
	Protein	MLVRLLYASRAVDTSPEAIEAILTQSRQHNPGS GITGVLCYGGGVFLQAIEGGRSAVSELYGHIQK DARHKNVELLHYEEISERRFGGWTMGQVNLS KINHSILLKYSEKPELDPYAVSGKVSFALLEEL MATASIIGRA
BNBA01000017.1/83078-82865 — unrelated fmn riboswitch sequence (negative control)		
	RNA	GAACGUCUUCAGGGCGGGGUGAGAUUCCCC ACCGGCGGUAGGUGCGGCGCCCUUGCGGCG AGGCACGAGCCCGCGAGCGCUUUCGAUGGC AUGCGAUCACGCCCCGCCGCGUGGGUCCGU GCCUUGGAAGGUCAGCAGAUCCGGUCAGAU GCCGGAGCCGACGGUCGGCAGGGCCCGGGG CCAUGCCGGAAAGUCCGGAUGAAAGAAGAC GGUA
	BLUF protein	MLVRLLYASRAVDTSPEAIEAILTQSRQHNPGS GITGVLCYGGGVFLQAIEGGRSAVSELYGHIQK DARHKNVELLHYEEISERRFGGWTMGQVNLS KINHSILLKYSEKPELDPYAVSGKVSFALLEEL MATASIIGRA

Table 5.7: Candidate clusters from the flagellin (IPR001029, PF00669) motif screen. Unlike the negative controls in Table **??**, this domain has a described precedent for associated *cis*-regulatory elements coupling flagellar gene expression to assembly state (discussed in Section **??**).

Cluster	Direction	Cov. pairs*	True	Sen. (%)	F-score
327	Downstream	11	40	27.50	43.14
545	Downstream	2	48	4.17	8.00
589	Downstream	4	103	3.88	7.48
16	Upstream	1	41	2.44	4.76
265	Downstream	1	36	2.78	5.41
725	Downstream	1	45	2.22	4.35
161	Downstream	1	77	1.30	2.56
272	Downstream	1	79	1.27	2.50
214	Upstream	1	165	0.61	1.20

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 in all cases.

5.3 Supplementary Figures

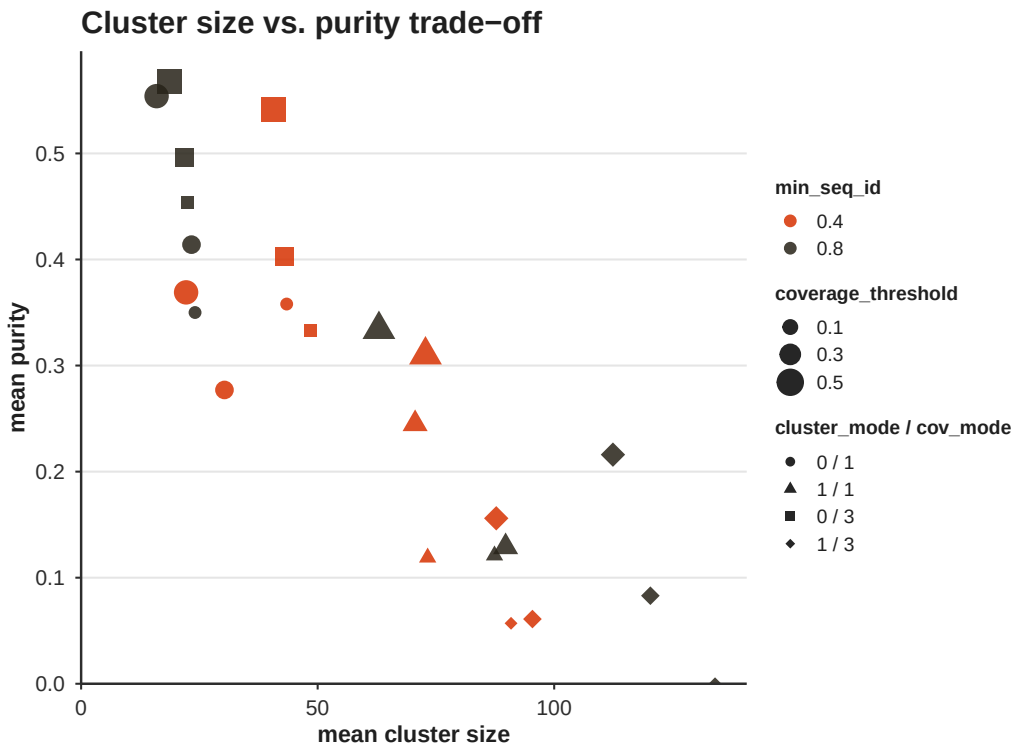


Figure 5.1: Clustering parameter selection for positive control pipeline. The clustering step of the positive control pipeline was run with a range of parameters, and the number of clusters and singleton-genome clusters were recorded.

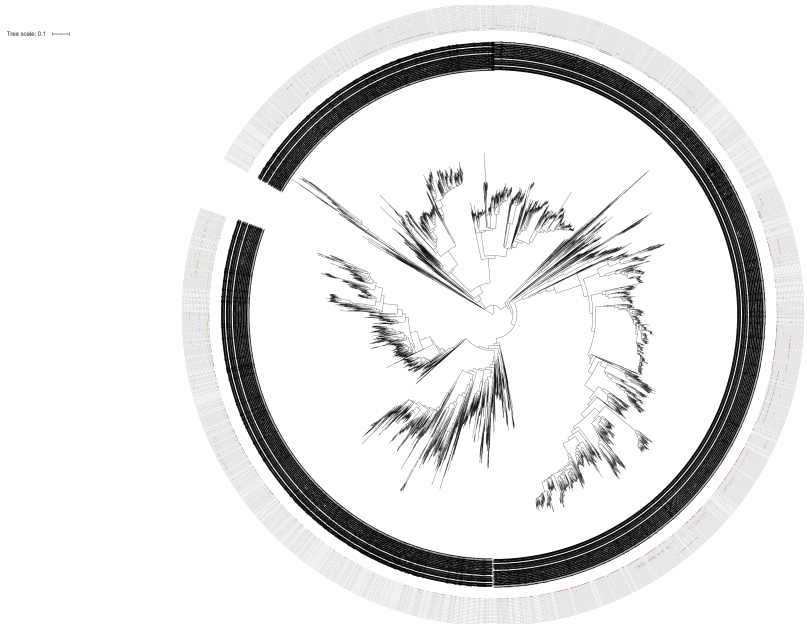


Figure 5.2: Prevalence of homologous protein sequences neighbouring FMN riboswitch. Top 10 clusters by presence per genome are shown around tree, in order of Appendix Table 5.4. Tree is annotated with analysis output of the positive control preperation pipeline output. It shows the GTDB species taxonomy of representative genomes containing the FMN riboswitch (as annotated by Rfam).

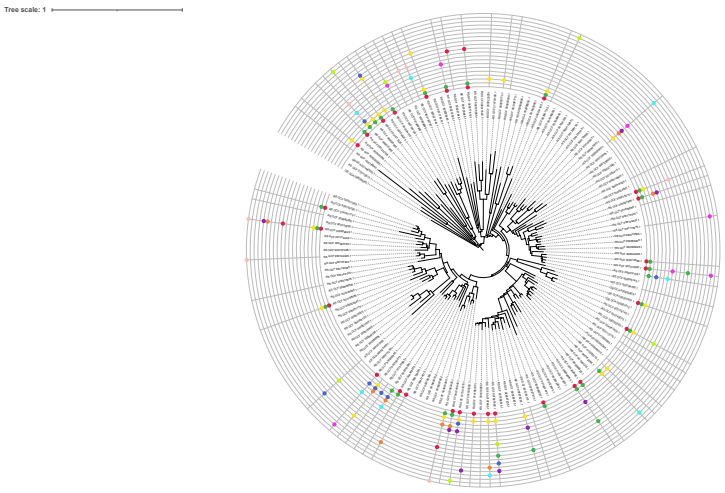
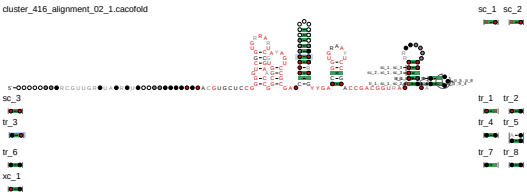
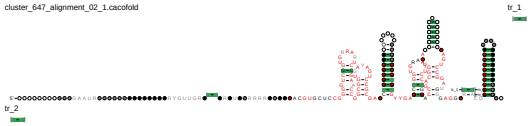


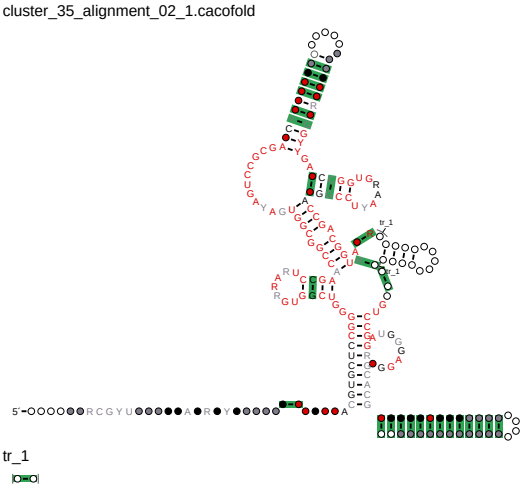
Figure 5.3: Prevalence of homologous protein sequences neighbouring CRISPR-Direct Repeat 42. Top 10 clusters by presence per genome are shown around tree, in order of Appendix Table 5.4. Tree is annotated with analysis output of the positive control preperation pipeline output. It shows the GTDB species taxonomy of representative genomes containing CRISPR-Direct Repeat 42 (as annotated by Rfam).



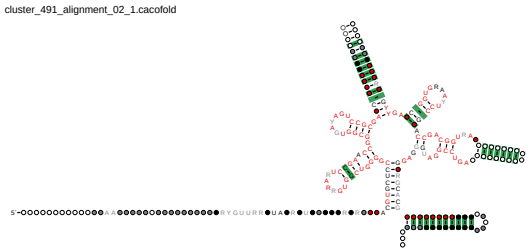
(a) Candidate 416 (F = 78.95)



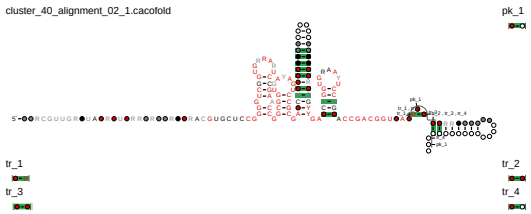
(b) Candidate 647 (F = 73.17)



(c) Candidate 35 (F = 62.07)

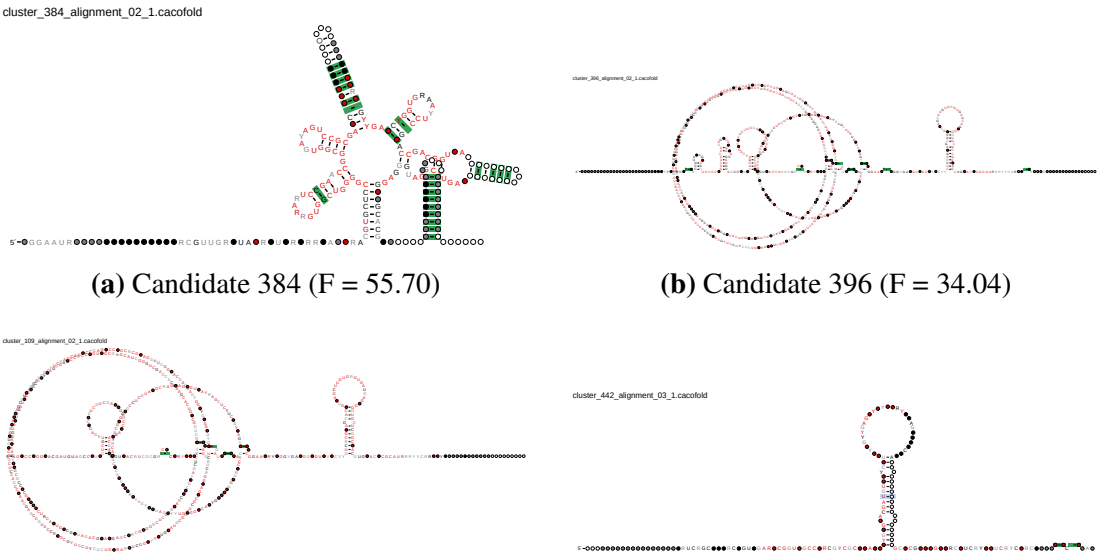


(d) Candidate 491 (F = 60.67)



(e) Candidate 40 (F = 56.67)

Figure 5.4: Predicted covarying secondary structures of remaining FMN riboswitch candidates (part 1 of 2). R-scape CaCoFold structures generated from iteratively expanded alignments, ordered by F-score.



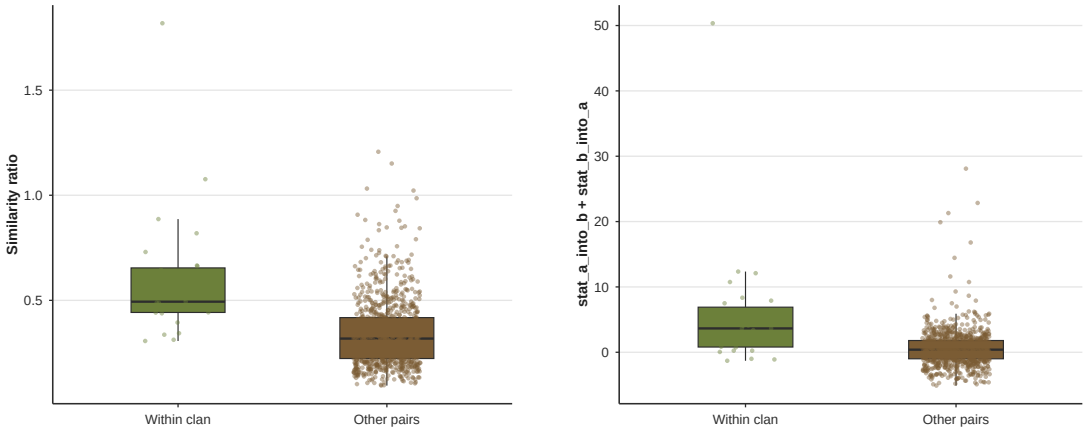
(a) Candidate 384 (F = 55.70)

(b) Candidate 396 (F = 34.04)

(c) Candidate 109 (F = 32.26)

(d) Candidate 442 (F = 20.00)

Figure 5.5: Predicted covarying secondary structures of remaining FMN riboswitch candidates (part 2 of 2). R-scape CaCoFold structures generated from iteratively expanded alignments, ordered by F-score. 5.5b and 5.5c correspond to the two low-identity outlier clusters discussed in the results text.



(a) Similarity ratio (minmax)

(b) Bitscore median vs. null

Figure 5.6: Within- vs. between-clan separation for the two initial classification approaches. (a): shows pairwise similarity ratio scores, motivated by common-ancestor theory but lacking a statistical null. **(b)**: shows median bitscore compared against a dinucleotide-shuffled bootstrap null distribution. Within-clan (n = 27) and other (n = 919) pairs are drawn from annotated Rfam entries.